Chapter 16 – Wave Motion

Propagation of a disturbance
Speed of waves in strings

Analysis Model: Traveling Wave
Reflection and Transmission

Waves

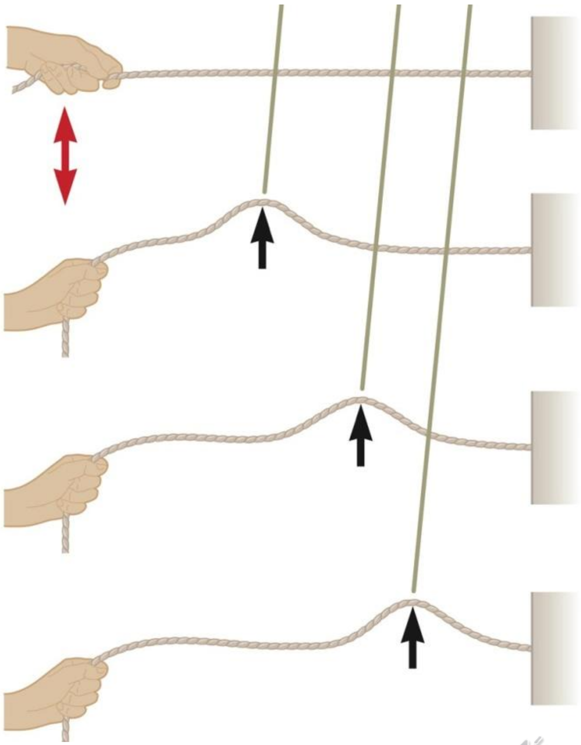
- **Central feature** to wave motion: energy is transferred over a distance, but matter is not.

- **Types of waves:**
  - Mechanical waves – water waves, sound waves and seismic waves. Governed by Newton's laws. Only exist in material mediums ($A << \lambda$).
  - Electromagnetic waves – infrared, visible, ultraviolet light, radio and TV waves, microwaves, x-rays and radar waves. Requires no material medium to exist.
  - ?? Matter waves ?? – e-, neutrons, etc

    Seems like MATTER IS TRANSFERRED!

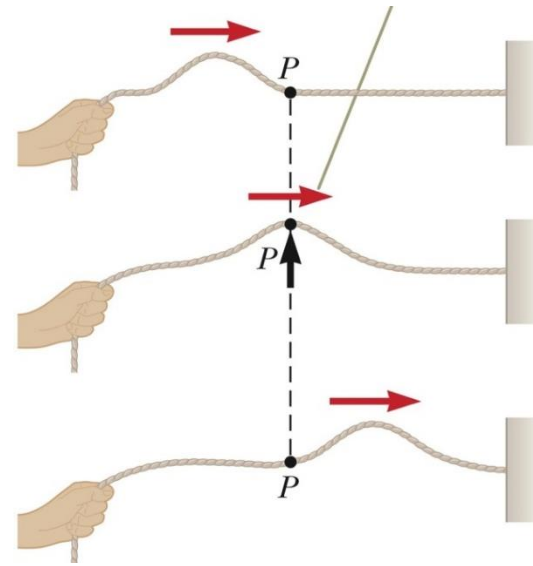
Propagation of a Disturbance

- All mechanical waves require:
  - Some **source of disturbance**
  - **Medium** containing elements that can be disturbed
  - A physical **mechanism** through which **elements** of the medium can influence each other.
- Flick one end of a taut string that is fixed at the other end.
- Pulse
  - Hand is the source of disturbance
  - String is the medium
  - Elements (constituents) of the string are connected to each other.

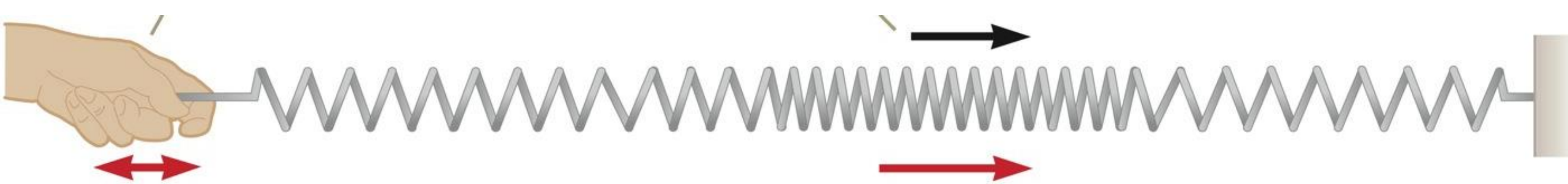


Propagation of a Disturbance

- Transverse Waves: Displacement of each element is perpendicular to the direction in which wave is traveling.

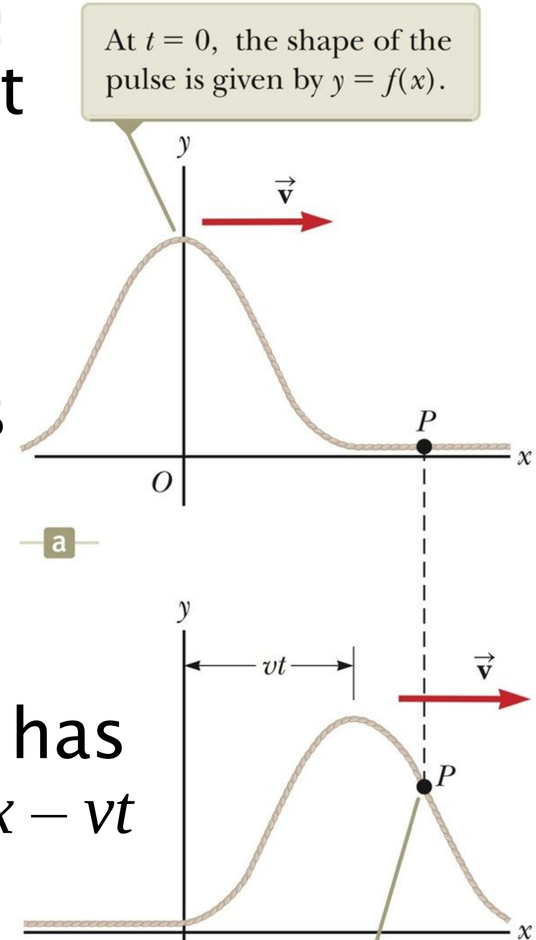


- Longitudinal Waves: Elements are displaced in the same direction as wave motion.



Propagation of a Disturbance

- Consider a Pulse travelling to the right along a string.
- Shape and position of pulse at $t = 0$ is $y(x,0) = f(x)$ (**Mathematical Description!**)
- Speed of the pulse $= v$. After time t has elapsed, the pulse has moved a distance vt.
- Shape of pulse **does not change.**
- An element of the string at x at time t has the same y position as an element at $x - vt$ had at $t = 0$. [whole x–axis minus vt]



$$y(x,t) = y(x - vt, 0) \text{ same y pos at } x = 0 \;\; \& \;\; x - vt$$

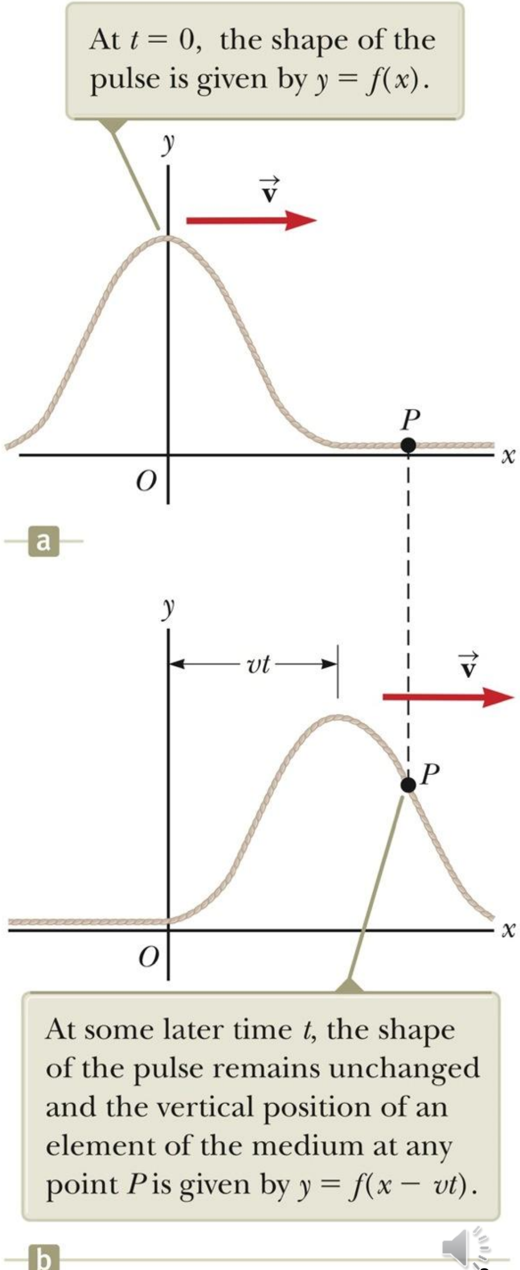
$$y(x,t) = f(x - vt) \text{ traveling to the right}$$

$$y(x,t) = f(x + vt) \text{ traveling to the left}$$

Propagation of a Disturbance

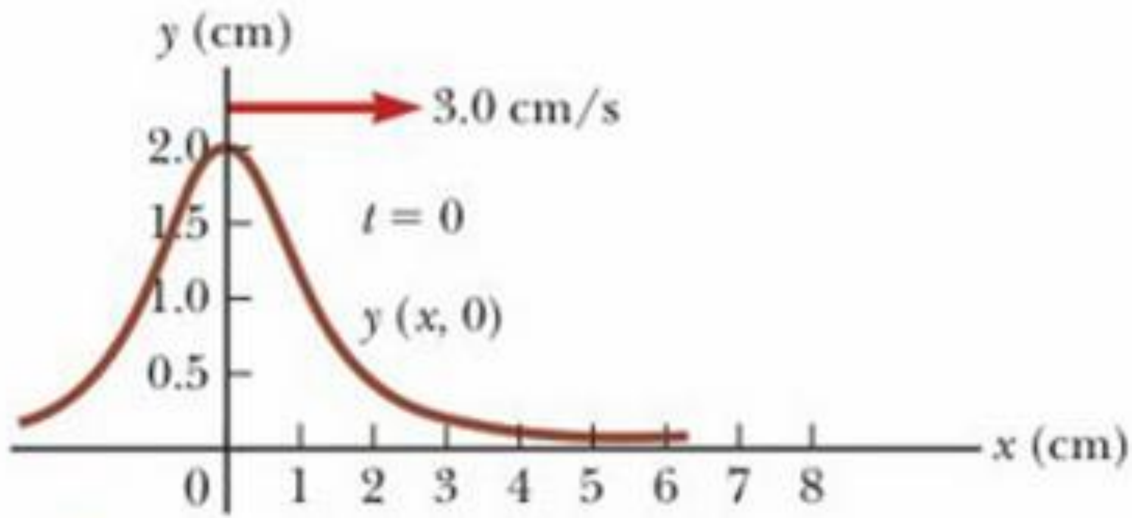
- $y(x,t)$ **is called a Wave function**:
- **Wave form**: Snapshot at a certain t.
- Consider particle at point P as time passes. Its displacement starts at $y = 0$, increases to a max, then returns to zero.

Speed of pulse is $v \rightarrow$ crest of pulse has traveled to right a distance vt at time t.



Example 16.1

- Work through the problem



$$\boxed{y(x, t) = \frac{2}{(x - 3.0t)^2 + 1}}$$

Example 16.1:
A Pulse Moving to the Right

A pulse moving to the right along the *x* axis is represented by the wave function

$$y(x,t) = \frac{2}{(x - 3.0t)^2 + 1}$$

where *x* and *y* are measured in centimeters and t is measured in seconds.

Find expressions for the wave function at $t = 0$, $t = 1.0$ s, and $t = 2.0$ s.

Example 16.1:
A Pulse Moving to the Right

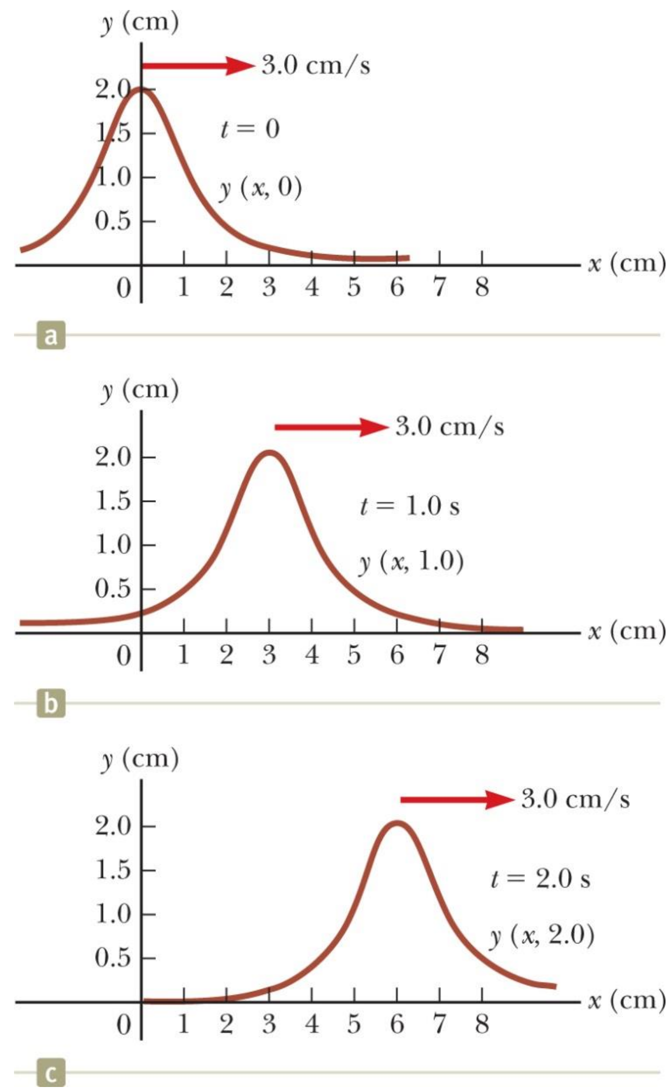
$$y(x,t) = \frac{2}{(x-3.0t)^2 + 1}$$

$$y(x,t) = f(x - vt)$$

$$y(x,0) = \boxed{\frac{2}{x^2 + 1}}$$

$$y(x,1.0) = \boxed{\frac{2}{(x-3.0)^2 + 1}}$$

$$y(x,2.0) = \boxed{\frac{2}{(x-6.0t)^2 + 1}}$$



Example 16.1:
A Pulse Moving to the Right

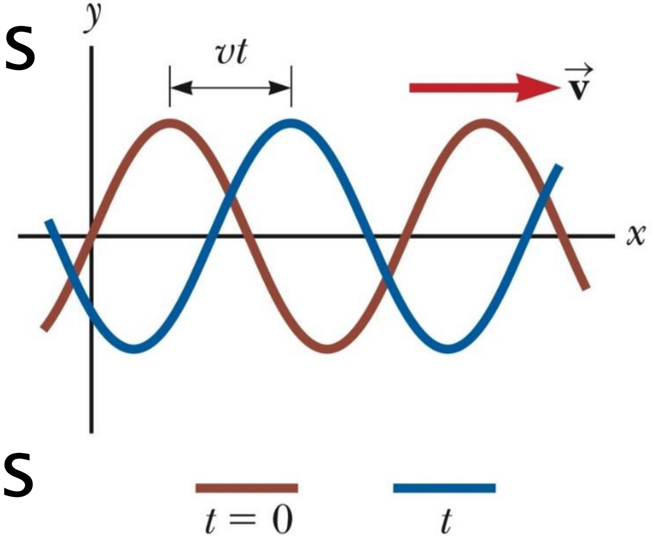
What if the wave function were

$$y(x,t) = \frac{4}{(x+3.0t)^2+1}$$

How would that change the situation?

Analysis model: Traveling Wave

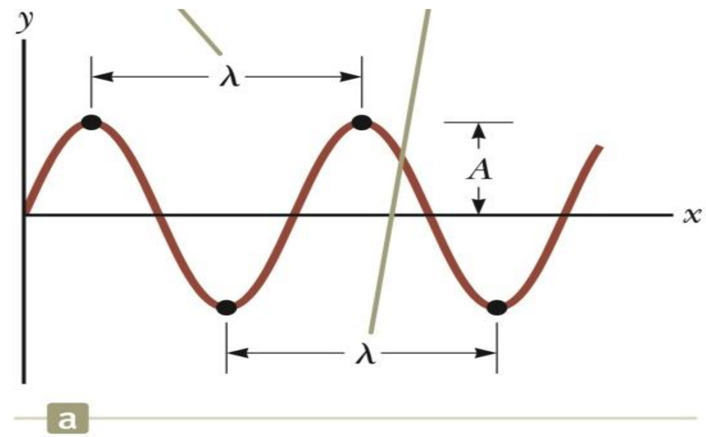
- Sinusoidal Wave: Can be set up in a string.
- Are important wave functions
- Brown curve at: $t = 0$
- Blue curve at: t
- Wave moves to the right
- If we focus on an element of the medium – element moves up and down



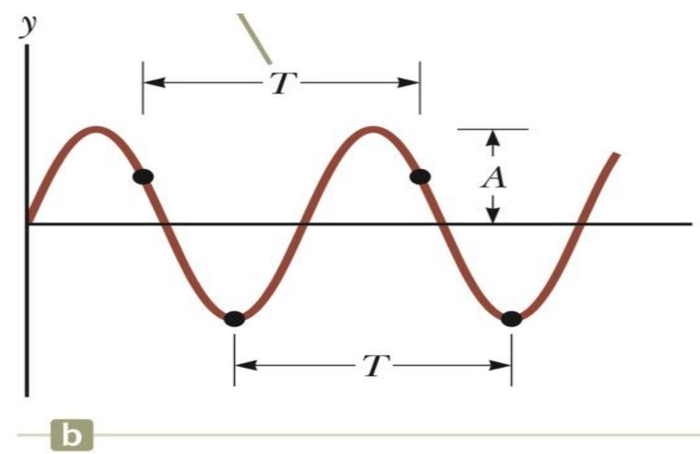
- It is important to differentiate between the motion of the wave and the motion of the elements of the medium.

Analysis model: Traveling Wave

- **Top**: Traveling wave through medium $\boxed{y \text{ vs } x}$
  - Crest
  - Trough
  - Wavelength – λ



- **Bottom figure**: position of an element as a function of time $\boxed{y \text{ vs } t}$
  - Period – T
  - Frequency – $f = \dfrac{1}{T}$
  - Amplitude – A



Analysis model: Travelling Wave

$$y(x,t) = A\sin(kx - \omega t)$$ *

- Amplitude A : maximum displacement from the position of rest. For transverse wave, transverse displacement y.
- phase $\equiv (kx - \omega t)$.
- Wavelength λ : distance between repetitions of the wave shape.
- Angular wave number: $k \equiv \frac{2\pi}{\lambda}$ (rad/m)
- Period: time between repetitions of the motion.
- Angular frequency (rad/s) $\omega \equiv \frac{2\pi}{T}$
- Frequency (Hz) : $f = \frac{1}{T} = \frac{\omega}{2\pi}$
- Wave speed:

$$v = \lambda f = \frac{\omega}{k}$$

Analysis model: Traveling Wave

- General form for a wave function

$$y(x,t) = A\sin(kx - \omega t + \phi)$$

- ϕ is the phase constant – takes into consideration when the wave does not start at $t = 0$.
- Can be determined by examining the initial conditions of the wave.

Mathematical Descriptions of Waves

$$v = \frac{\Delta x}{\Delta t} = \frac{\lambda}{T} = \lambda f$$

$$\Rightarrow y(x,t) = A\sin(kx - \omega t)$$

$$\Rightarrow y(x,t) = A\sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right]$$

$$\boxed{k \equiv \frac{2\pi}{\lambda}} \text{ and } \boxed{\omega \equiv \frac{2\pi}{T} = 2\pi f}$$

$$\boxed{y(x,t) = A\sin(kx - \omega t)}$$

$$v = \frac{\omega}{k} \text{ and } \boxed{v = \lambda f}$$

$$y(x,t) = A\sin(kx - \omega t + \phi)$$

Quick Quiz 16.2 Part I

A sinusoidal wave of frequency *f* is traveling along a stretched string. The string is brought to rest, and a second traveling wave of frequency 2*f* is established on the string.

What is the wave speed of the second wave?

(a) twice that of the first wave
(b) half that of the first wave
(c) the same as that of the first wave
(d) impossible to determine

Quick Quiz 16.2 Part I

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(d) impossible to determine

Quick Quiz 16.2 Part II

A sinusoidal wave of frequency f is traveling along a stretched string. The string is brought to rest, and a second traveling wave of frequency $2f$ is established on the string.

What is the wavelength of the second wave?

(a) twice that of the first wave
(b) half that of the first wave
(c) the same as that of the first wave
(d) impossible to determine

Quick Quiz 16.2 Part II

A sinusoidal wave of frequency *f* is traveling along a stretched string. The string is brought to rest, and a second traveling wave of frequency 2*f* is established on the string.

What is the wavelength of the second wave?

(a) twice that of the first wave

(b) half that of the first wave

(c) the same as that of the first wave

(d) impossible to determine

Quick Quiz 16.2 Part III

A sinusoidal wave of frequency f is traveling along a stretched string. The string is brought to rest, and a second traveling wave of frequency $2f$ is established on the string.

What is the amplitude of the second wave?

(a) twice that of the first wave
(b) half that of the first wave
(c) the same as that of the first wave
(d) impossible to determine

Quick Quiz 16.2 Part III

A sinusoidal wave of frequency f is traveling along a stretched string. The string is brought to rest, and a second traveling wave of frequency $2f$ is established on the string.

What is the amplitude of the second wave?

(a) twice that of the first wave

(b) half that of the first wave

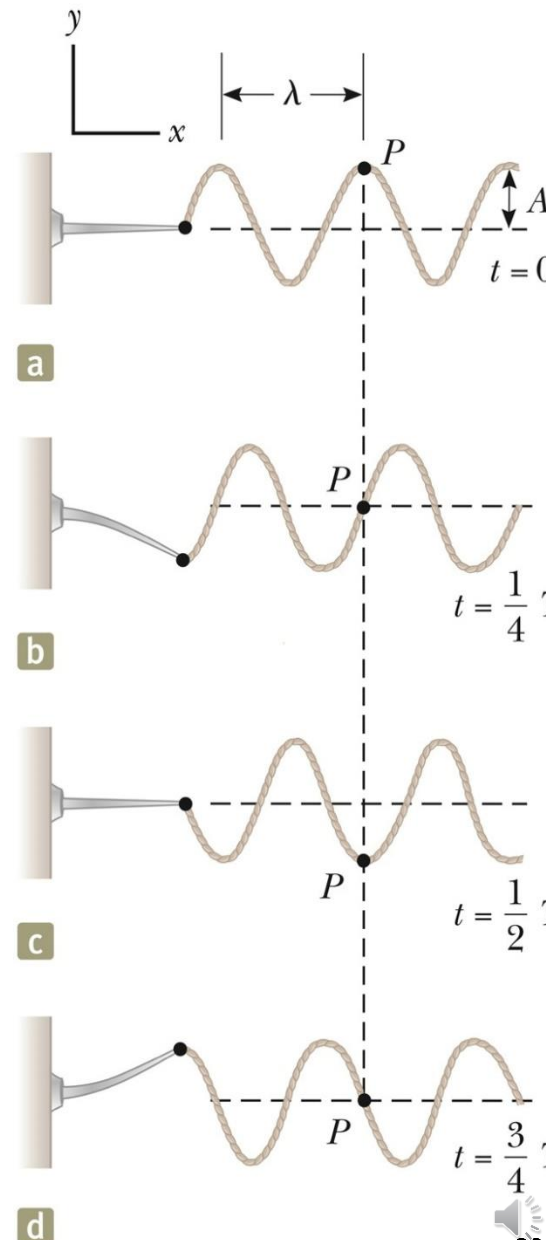
(c) the same as that of the first wave

(d) impossible to determine

Sinusoidal Waves on Strings

- A series of pulses vs 1 pulse
- Leads to A Wave
- Arm generating pulses = SHM
- Fig shows Snapshots of the wave at intervals of $T/4$
- Each element of string – SHM
- BUT wave moves to the right $+x$ direction with a speed v.
- At $t=0$ Snapshot/waveform @ (a):

$$y(x,t) = A\sin(kx - \omega t)$$



Sinusoidal Waves on Strings

$$y(x,t) = A\sin(kx - \omega t)$$

- This expression describes motion of an element on the string.
- Element *P* "fixed" at x–coordinate.
- **Transverse speed** (not **wave speed**):

$$v_y = \left.\frac{dy}{dt}\right|_{x=\text{constant}} = \frac{\partial y}{\partial t} = -\omega A\cos(kx - \omega t)$$

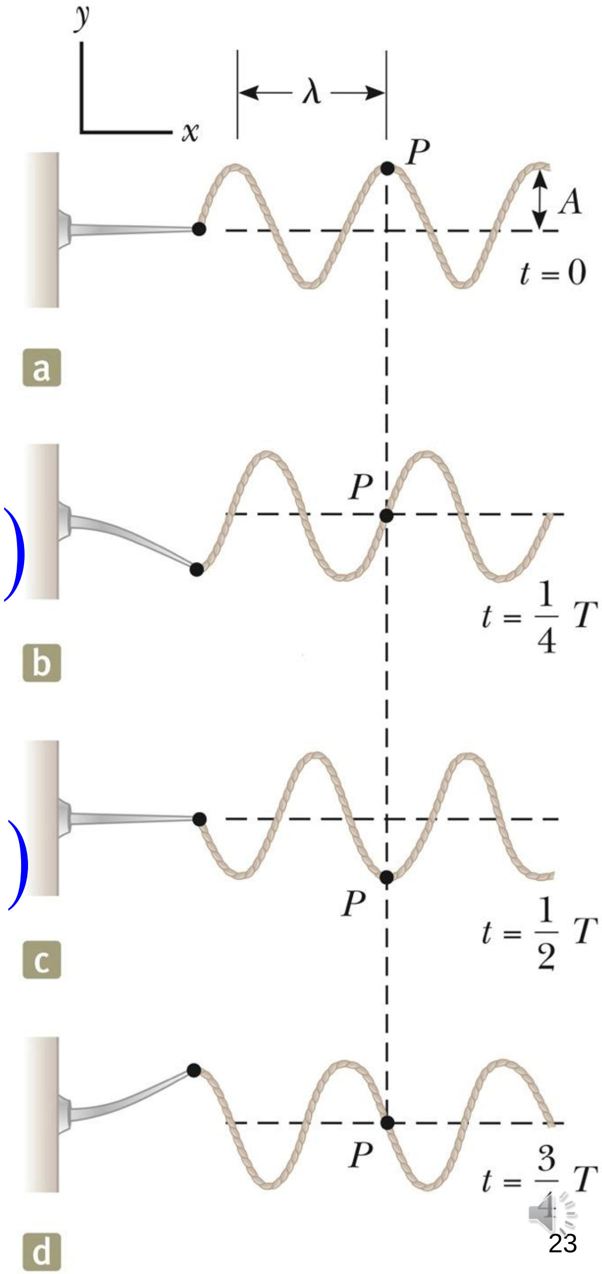
- **Transverse acceleration:**

$$a_y = \left.\frac{dv_y}{dt}\right|_{x=\text{constant}} = \frac{\partial v_y}{\partial t} = -\omega^2 A\sin(kx - \omega t)$$

-

$$v_{y,\max} = \omega A \qquad a_{y,\max} = \omega^2 A$$

Where do the maxima occur?



The Speed of Waves on Strings

Assume travelling inertial frame, unif. linear mass density
Assume Amplitude << wavelength. Pulse to RHS @ **v**
Assume Pulse doesn't affect uniform tension, *T*

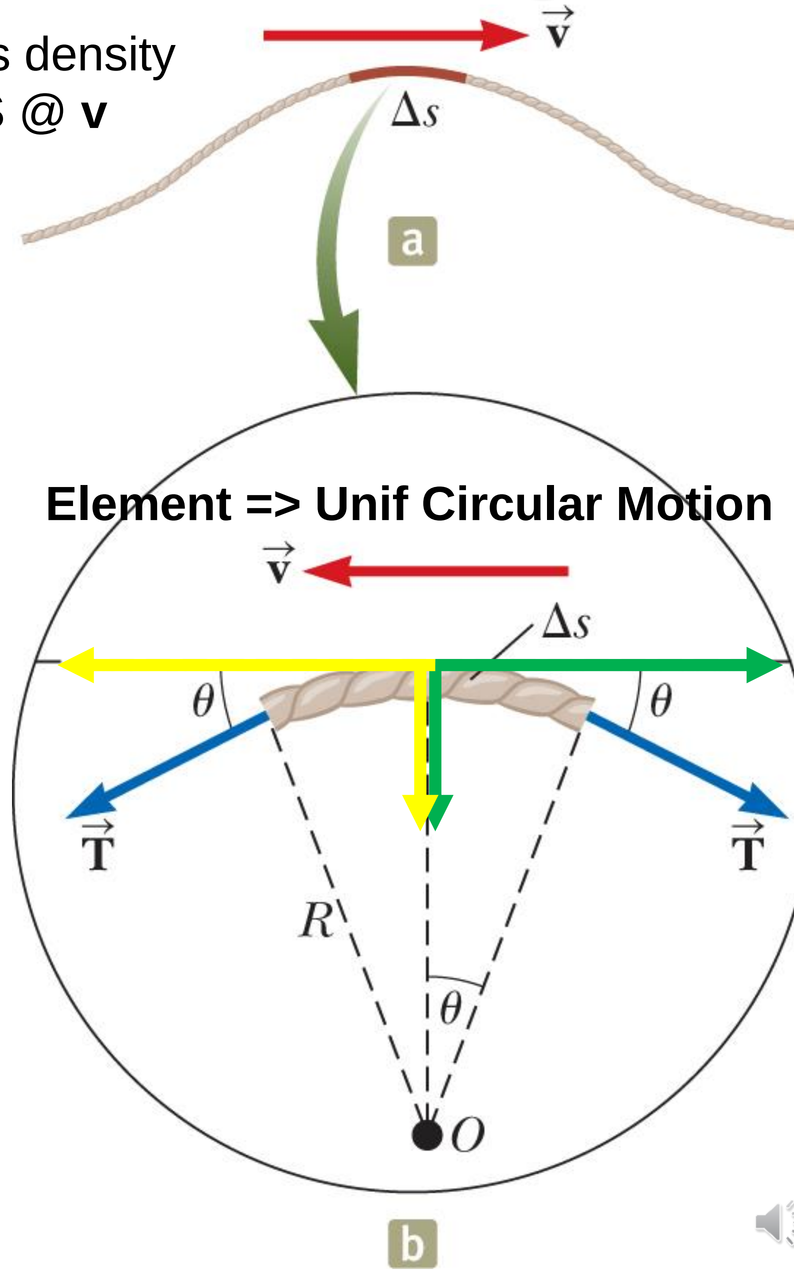
$$F_r = 2T\sin\theta \approx 2T\theta$$

$$\Delta m = \mu\Delta s = \mu R(2\theta)$$

$$F_r = \frac{\Delta m\ v^2}{R} \Rightarrow 2T\theta = \frac{2\mu R\theta v^2}{R}$$

$$\Rightarrow T = \mu v^2$$

$$\boxed{v = \sqrt{\frac{T}{\mu}}}$$

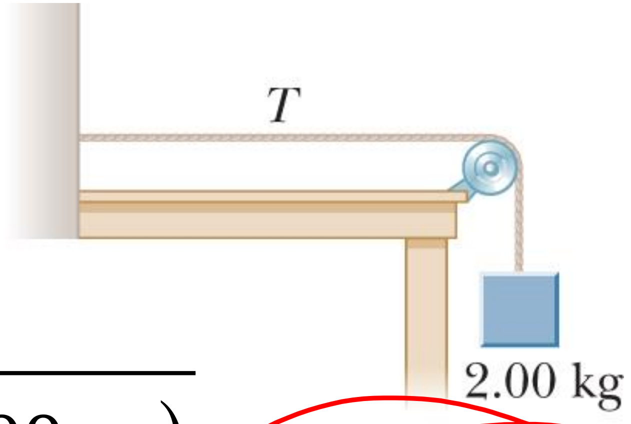


Example 16.3:
The Speed of a Pulse on a Cord

A uniform string has a mass of 0.300 kg and a length of 6.00 m. The string passes over a pulley and supports a 2.00-kg object. Find the speed of a pulse traveling along this string.

$$\sum F_y = T - m_{\text{block}}\, g = 0 \;\Rightarrow T = m_{\text{block}}\, g$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{m_{\text{block}}\, g \ell}{m_{\text{string}}}}$$

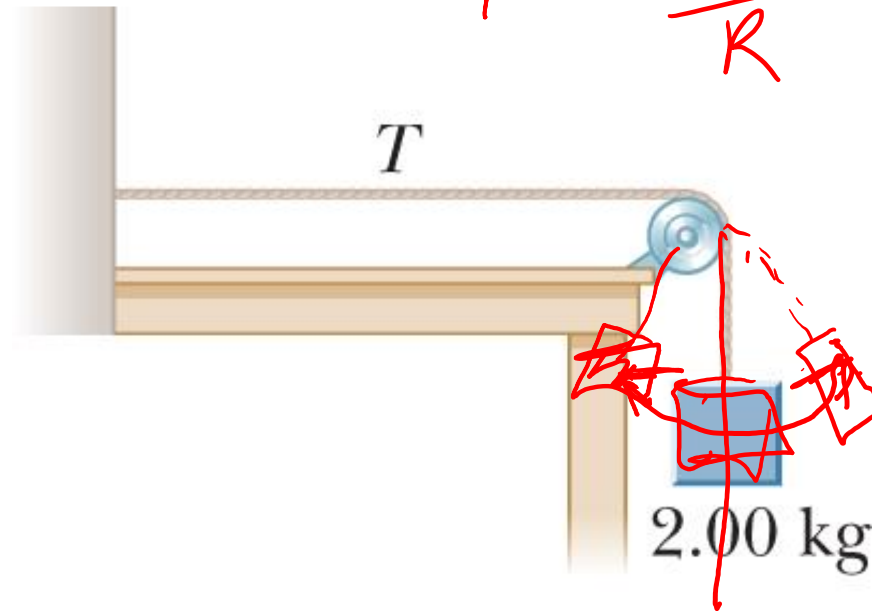


$$v = \sqrt{\frac{(2.00\text{ kg})(9.80\text{ m/s}^2)(6.00\text{ m})}{0.300\text{ kg}}} = \boxed{19.8\text{ m/s}}$$

Example 16.3:
The Speed of a Pulse on a Cord

What if the block were swinging back and forth with respect to the vertical like a pendulum? How would that affect the wave speed on the string?

$\Sigma F = \frac{mv^2}{R}$



$$\sum F_y = T - m_{\text{block}}\, g = m_{\text{block}} \frac{v^2}{r}$$

$$\therefore T = \boxed{m_{\text{block}} \left(\frac{v^2}{r} + g\right)}$$

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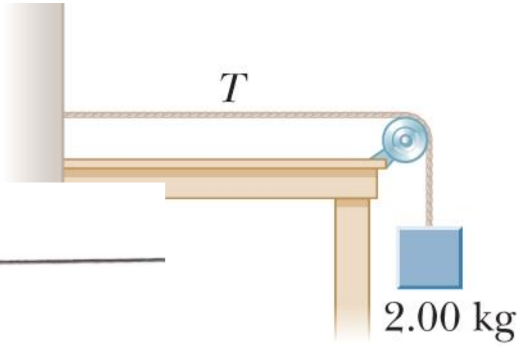

$$\sum F_y = T - m_{\text{block}}\, g = m_{\text{block}} \frac{v^2}{r}$$

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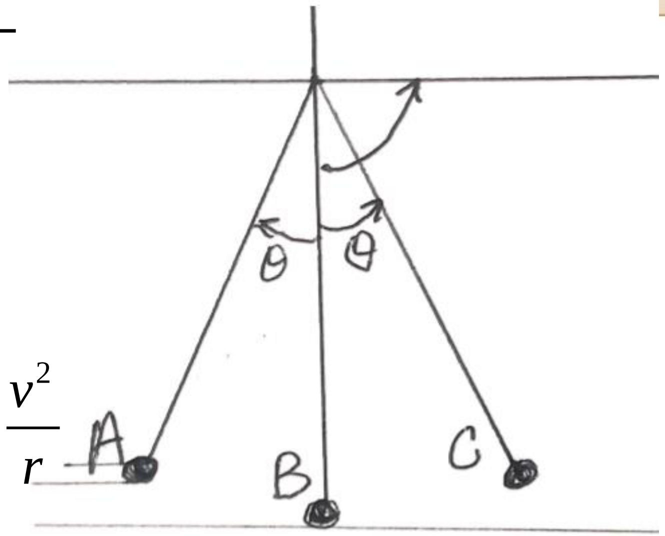

$$\sum F_r = T - m_{\text{block}}\, g = m_{\text{block}} \frac{v^2}{r}$$

$$\therefore T = \left(\cancel{m_{\text{block}} \frac{v^2}{r}} + m_{\text{block}}\, g \cos\theta\right)$$

$$\sum F_y = T - m_{\text{block}}\, g = m_{\text{block}} \frac{v^2}{r}$$

$$\therefore T = m_{\text{block}} \left(\frac{v^2}{r} + g\right)$$

Example 16.3:
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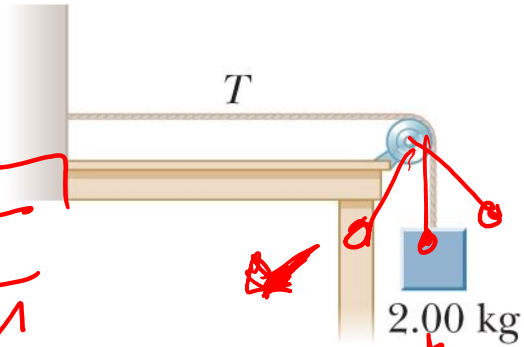


$$v = \sqrt{\frac{T}{\mu}}$$

$$\sum F_y = T - m_{\text{block}}\, g = m_{\text{block}} \frac{v^2}{r}$$

$$\therefore T = m_{\text{block}} \left(\frac{v^2}{r} + g\right)$$

The swinging block is categorized as a *particle under a net force*. The magnitude of one of the forces on the block is the tension in the string, which determines the wave speed. As the block swings, the tension changes, so the wave speed changes. When the block is at the bottom of the swing, the string is vertical and the tension is larger than the weight of the block because the net force must be upward to provide the centripetal acceleration of the block. Therefore, the wave speed must be greater than 19.8 m/s. When the block is at its highest point at the end of a swing, it is momentarily at rest, so there is no centripetal acceleration at that instant. The block is a particle in equilibrium in the radial direction. The tension is balanced by a component of the gravitational force on the block. Therefore, the tension is smaller than the weight and the wave speed is less than 19.8 m/s. With what frequency does the speed of the wave vary? Is it the same frequency as the pendulum?

Example 16.4: Rescuing the Hiker

$\Sigma F_y = ma$

An 80.0-kg hiker is trapped on a mountain ledge following a storm. A helicopter rescues the hiker by hovering above him and lowering a cable to him. The mass of the cable is 8.00 kg, and its length is 15.0 m. A sling of mass 70.0 kg is attached to the end of the cable. The hiker attaches himself to the sling, and the helicopter then accelerates upward. Terrified by hanging from the cable in midair, the hiker tries to signal the pilot by sending transverse pulses up the cable. A pulse takes 0.250 s to travel the length of the cable. What is the acceleration of the helicopter? Assume the tension in the cable is uniform.

Example 16.4: Rescuing the Hiker

$$v = \sqrt{\frac{T}{\mu}} \Rightarrow T = \mu v^2$$

$$\sum F = ma \Rightarrow T - mg = ma$$

$$a = \frac{T}{m} - g = \frac{\mu v^2}{m} - g = \frac{m_{\text{cable}} v^2}{\ell_{\text{cable}} m} - g = \frac{m_{\text{cable}}}{\ell_{\text{cable}} m}\left(\frac{\Delta x}{\Delta t}\right)^2 - g$$

$$a = \frac{(8.00\ \text{kg})}{(15.0\ \text{m})(150.0\ \text{kg})}\left(\frac{15.0\ \text{m}}{0.250\ \text{s}}\right)^2 - 9.80\ \text{m/s}^2 = \boxed{3.00\ \text{m/s}^2}$$

Problem

Given 3 equations (wave functions) for 3 waves:

$$(1) \quad y(x,t) = 2\sin(4x - 2t)$$

$$(2) \quad y(x,t) = \sin(3x - 4t)$$

$$(3) \quad y(x,t) = 2\sin(3x - 3t)$$

Q: Rank the waves according to their (a) wave speed and (b) maximum speed perpendicular to the waves' direction of travel (transverse speed).

$$v = \frac{\omega}{k}$$

$$v_{\text{trans}} = \frac{dy}{dt} = \omega y_m \cos(kx - \omega t)$$

$$= v_m \cos(kx - \omega t)$$

Problem

Here are the equations for 3 waves:

$$(1)\quad y(x,t) = 2\sin(4x - 2t)$$

$$(2)\quad y(x,t) = \sin(3x - 4t)$$

$$(3)\quad y(x,t) = 2\sin(3x - 3t)$$

Rank the waves according to their (a) wave speed and (b) maximum speed perpendicular to the waves direction of travel (transverse speed).

$$v = \frac{\omega}{k}$$

$$v_1 = \frac{2}{4} = ½$$

$$v_3 = \frac{3}{3} = 1$$

$$v_2 = \frac{4}{3} = 1.33$$

$$v_{\text{trans}} = \frac{dy}{dt} = \omega y_m \cos(kx - \omega t)$$

$$= v_m \cos(kx - \omega t)$$

$$\omega_1 y_m = 4$$

$$\omega_2 y_m = 4$$

$$\omega_3 y_m = 6$$

Problem 14

The equation of a transverse wave on a string is

$$y = (2.0\ \text{mm})\sin\left[\left(20\ \text{m}^{-1}\right)x - \left(600\ \text{s}^{-1}\right)t\right]$$

What is the amplitude, the frequency, the wavelength and the velocity of the wave?

Problem $y = A \sin(kx - \omega t)$

- The equation of a transverse wave on a string is

$$y = (2.0\ \text{mm})\sin\left[\left(20\ \text{m}^{-1}\right)x - \left(600\ \text{s}^{-1}\right)t\right]$$

What is the amplitude, the frequency, the wavelength and the velocity of the wave?

$$A = 2{,}0\ \text{mm}$$

$$f = \frac{\omega}{2\pi} = \frac{600}{2\pi} = 95.5\ \text{Hz}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{20} = 0.31\ \text{m}$$

$$v = \frac{\omega}{k} = \frac{600}{20} = 30\ \text{m/s}$$

or $v = \dfrac{\lambda}{T} = \dfrac{0.31}{(95.5)^{-1}} = 30\ \text{m/s}$

Problem

$y(t) = A\cos(kx - \omega t)$

The waves in a microwave oven are described by $\cos\left(50.3x - 15.1\times10^{18}\ t\right)$ with t measured in ns and x in m.

Show that f = 2.4 GHz, λ = 12.5 cm and v = 3.0 × 10^8 m/s.

Problem

- The waves in a microwave oven are described by $\cos\left(50.3x - 15.1\times10^{18}\, t\right)$ with t measured in ns and x in m. Show that f = 2.4 GHz, λ = 12.5 cm and v = 3.0 × 10^8 m/s.

$$f = \frac{\omega}{2\pi} = \frac{15.1\times10^{9}}{2\pi} = 2.4\times10^{9}\ \text{Hz} = 2.4\ \text{GHz}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{50.3\ \text{m}^{-1}} = 0.125\ \text{m} = 12.5\ \text{cm}$$

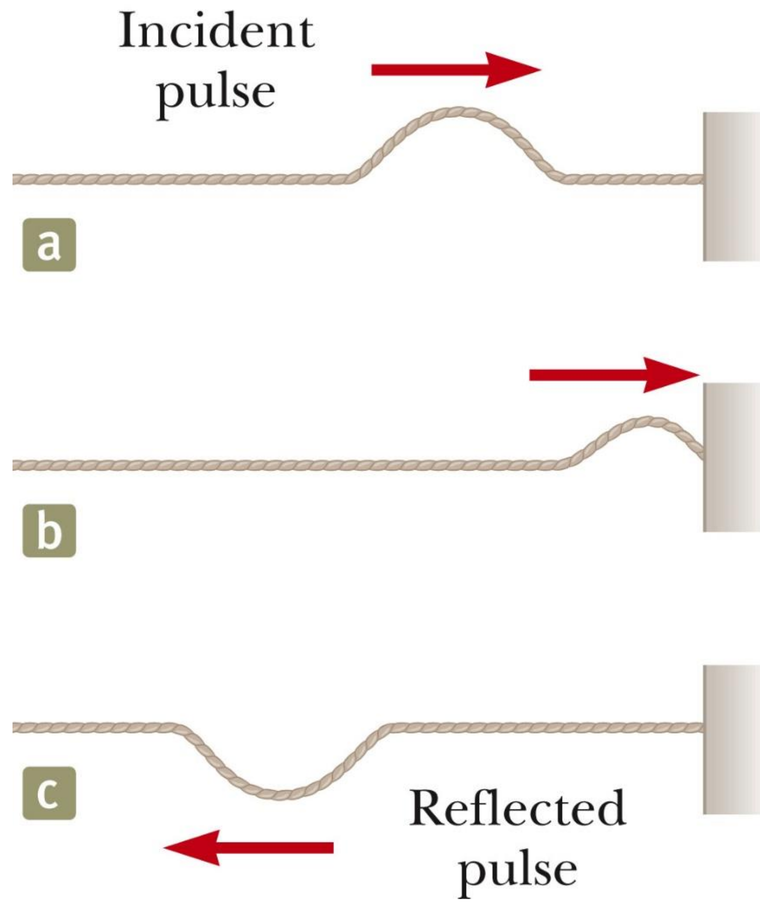
$$v = \frac{\omega}{k} = \frac{15.1\times10^{9}}{50.3} = 3.00\times10^{8}\ \text{m/s}$$

$$\text{or}\quad v = \frac{\lambda}{T} = \frac{0.125}{\left(2.4\times10^{9}\right)^{-1}} = 3.0\times10^{8}\ \text{m/s}$$

Reflection and Transmission- Self-Study

Reflection: Fixed end

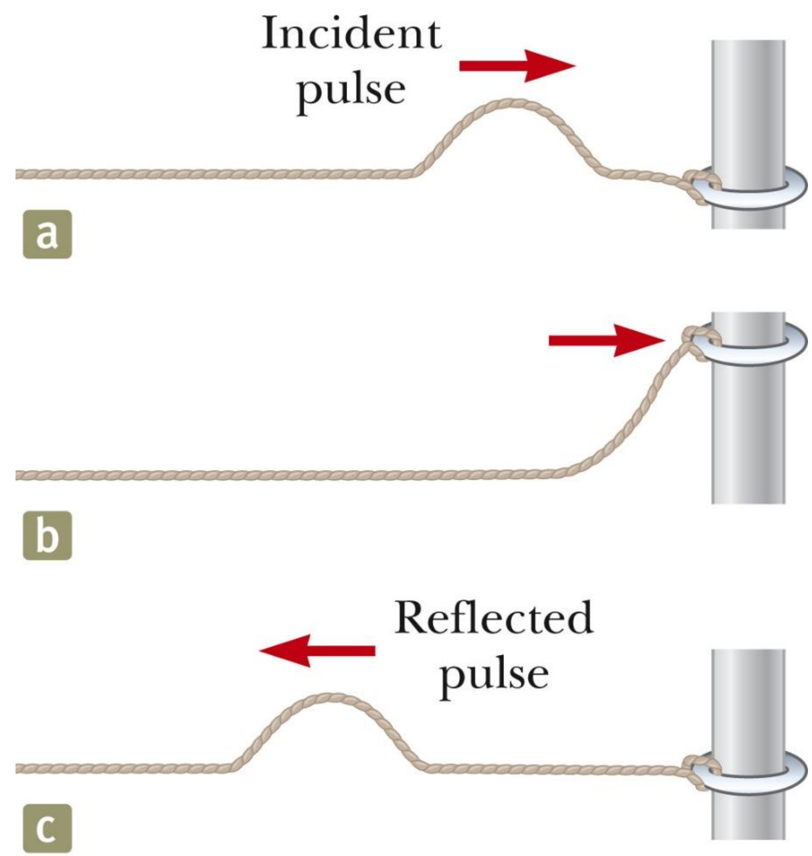
- When the pulse reaches the support, the pulse moves back along the string in the opposite direction.
- Reflection of the pulse.
- The pulse is inverted.
- Newton's 3rd law
  - When the pulse reaches the fixed end of the string, the string produces an upward force on the support.
  - The support must exert an equal-magnitude and oppositely directed reaction force on the string.



Reflection and Transmission

Reflection: Free end

- The string is free to move vertically.
- The pulse is reflected.
- The pulse is not inverted.
- The reflected pulse has the same amplitude as the initial pulse.



Reflection and Transmission

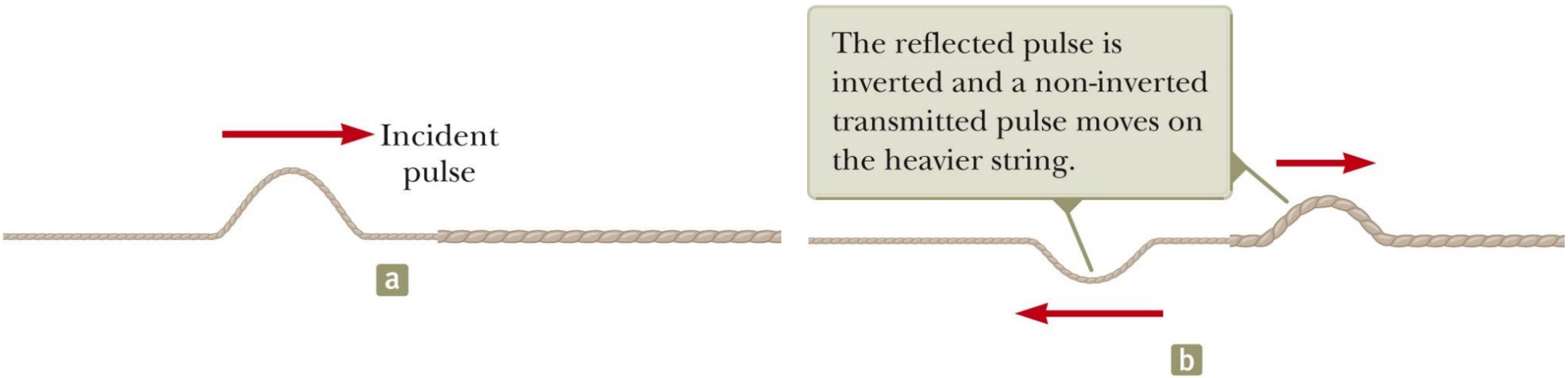
Transmission of a wave

- When the boundary is intermediate between the last two extremes.
- Part of the energy in the incident pulse is reflected and part undergoes **transmission.**
- Some energy passes through the boundary.

Reflection and Transmission

Transmission of a wave

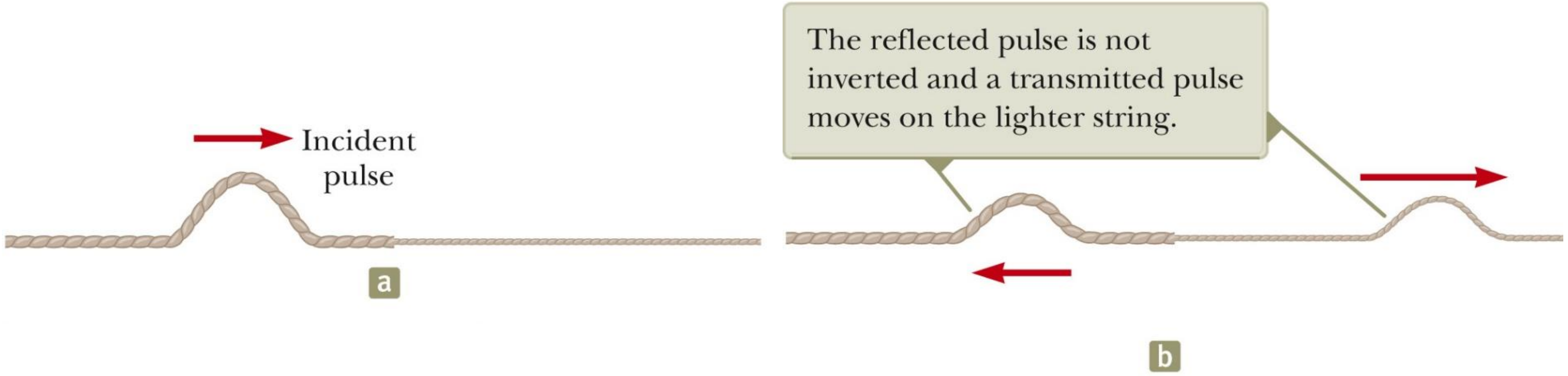
- When a light string is attached to a heavier string.



Reflection and Transmission

Transmission of a wave

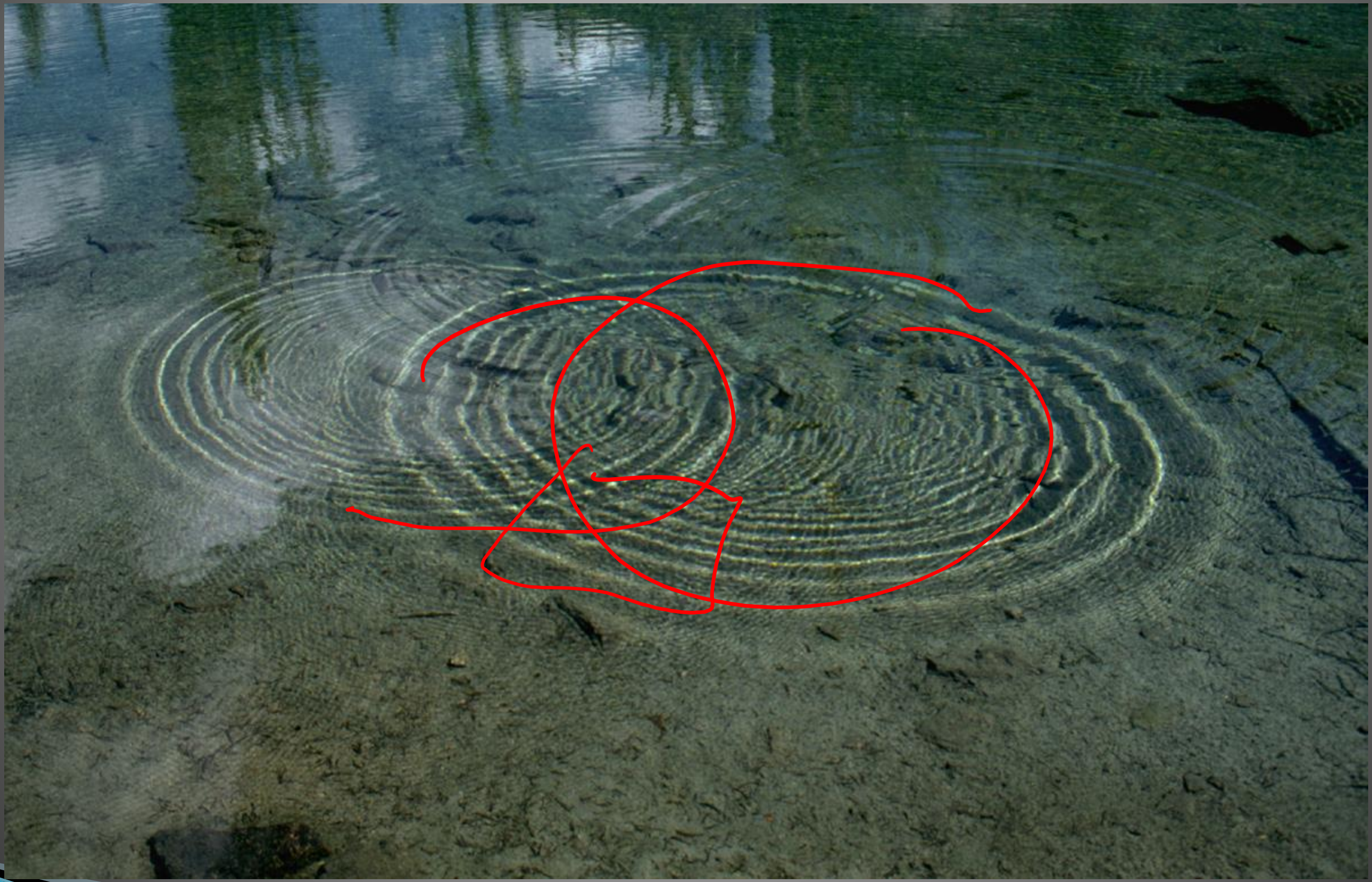
- When a heavy string is attached to a lighter string.



Chapter – Superposition and standing waves

Waves in interference
Waves under Boundary conditions
Standing waves

Superposition / Interference

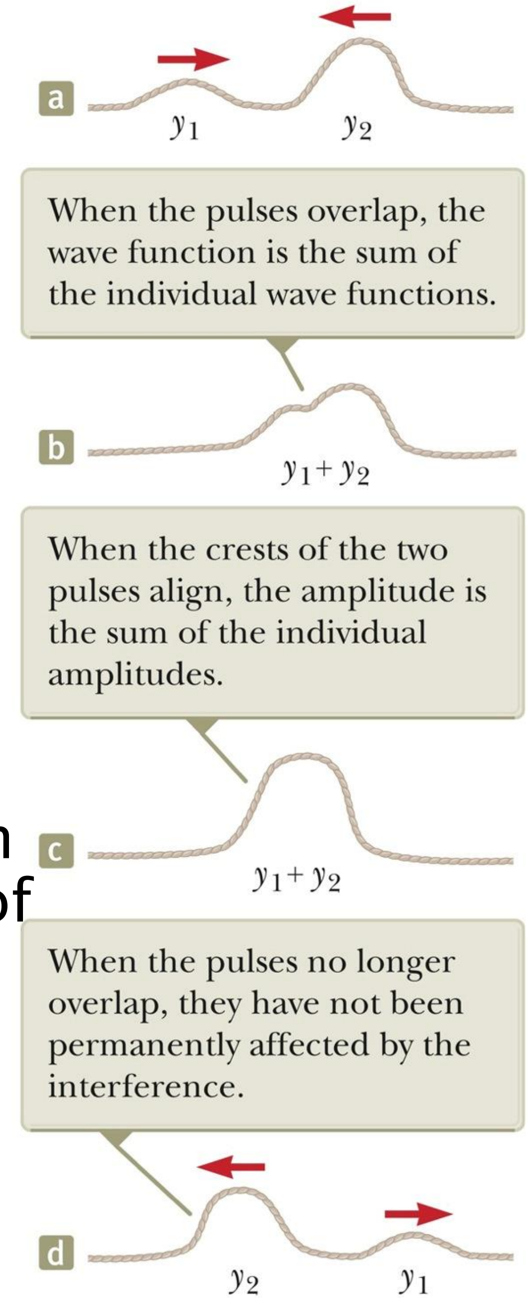


Superposition

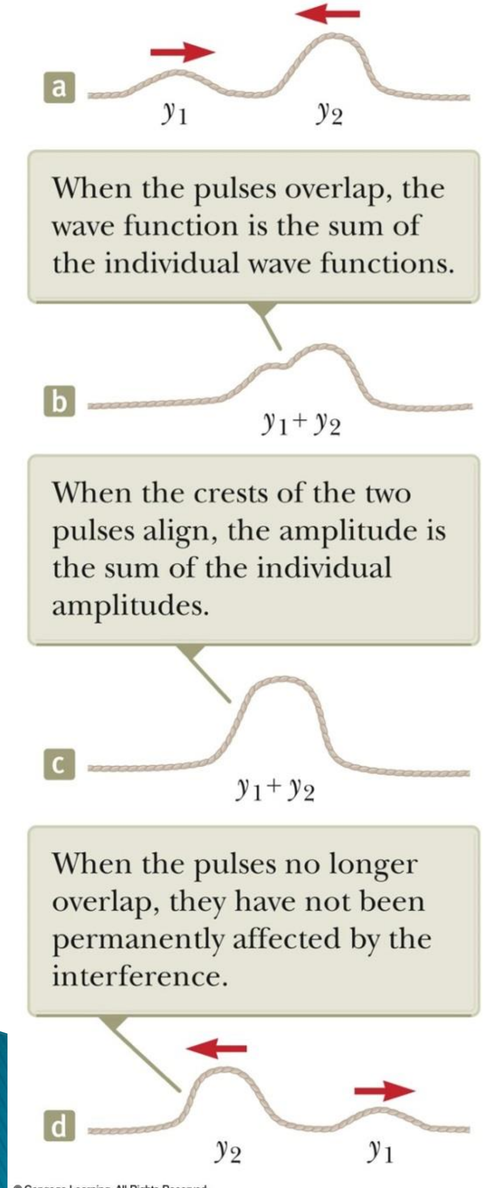
- Waves vs particles – waves can be at the same position at the same time, not particles.
- Consider 2 waves travelling along a string with displacements $y_1(x,t)$ and $y_2(x,t)$.
- When the waves overlap, the displacement is the algebraic sum of the individual displacements:

$$\boxed{y'(x,t) = y_1(x,t) + y_2(x,t).}$$

- Principle of Superposition – 2 or more travelling waves moving through a medium, can pass through the same point resulting in a wave function which is the algebraic sum of the individual waves e.g. listening to live music or stereo sound.
- Overlapping waves do not alter the travel of the waves.



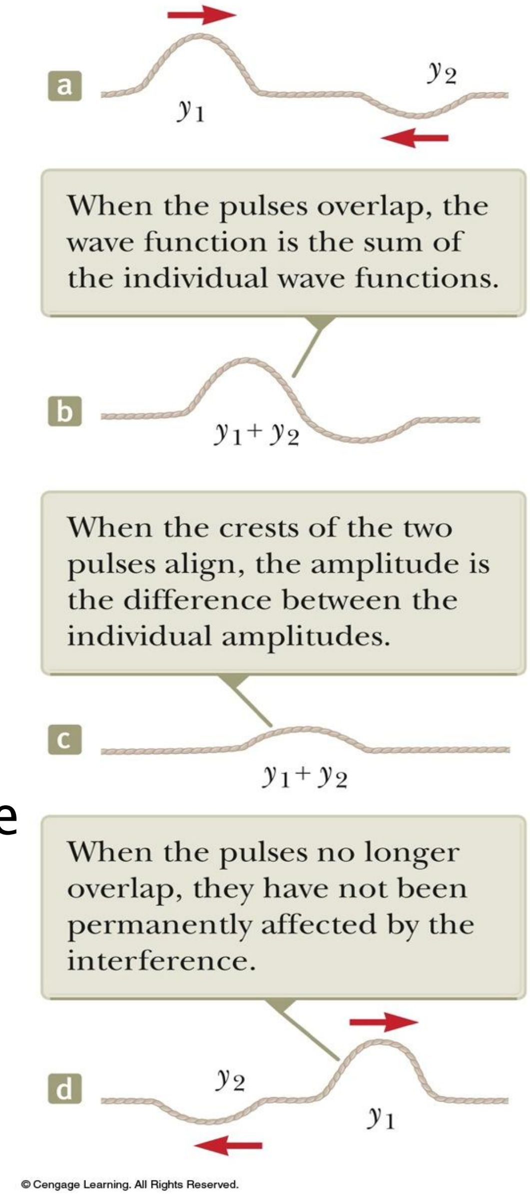
Superposition



Interference – combination of separate waves in the same region

Constructive interference

Destructive interference



Superposition of Sinusoidal waves

- Superposition allows us to add sinusoidal waves of different amplitudes, wavelengths and frequencies.
- The sum of these waves form an interference pattern.
- Consider **a wave**: $$y_1(x,t) = A\sin\left(kx - \omega t\right)$$
- And **another wave**: $$y_2(x,t) = A\sin\left(kx - \omega t + \phi\right)$$
- They have the same ω (and $\therefore f$), the same k (and $\therefore \lambda$) and the same A.
- They are out of phase by ϕ. (ie, they have a phase difference of ϕ)

Superposition of Sinusoidal waves

$$y'(x,t) = y_1(x,t) + y_2(x,t)$$

$$= A\sin\left(kx - \omega t\right) + A\sin\left(kx - \omega t + \phi\right)$$

Since $\sin\alpha + \sin\beta = 2\sin\frac{(\alpha+\beta)}{2}\cos\frac{(\alpha-\beta)}{2}$

$$\therefore \quad y'(x,t) = \boxed{\left[2A\cos\frac{\phi}{2}\right]}\sin\left(kx - \omega t + \frac{\phi}{2}\right)$$

- **Resultant**
  - Is also sinusoidal
  - Has same frequency and wavelength as original individual waves
  - Amplitude: $2A\cos(\phi/2)$

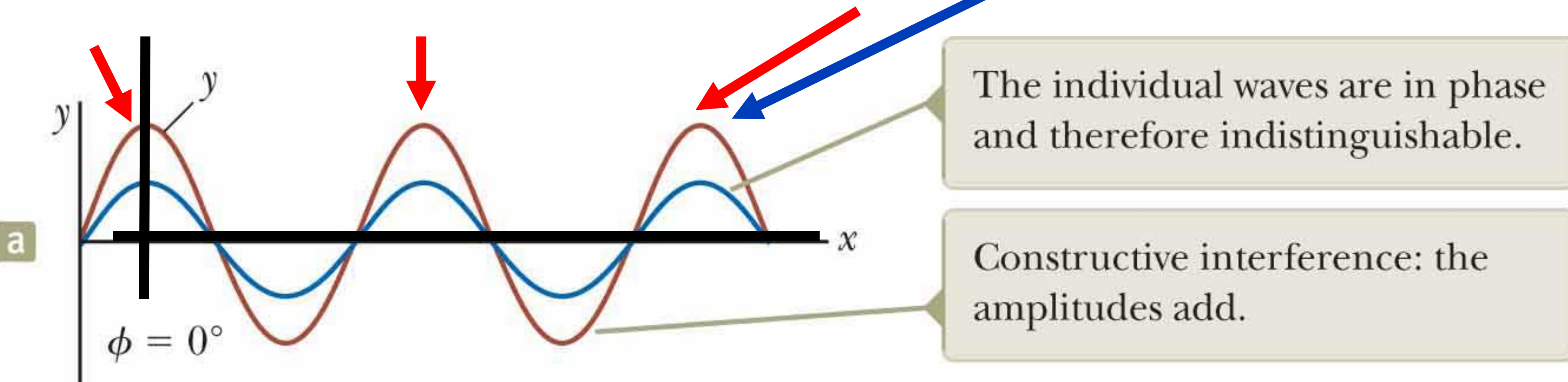
Superposition of Sinusoidal Waves

$$y = 2A\cos\left(\frac{\phi}{2}\right)\sin\left(kx - \omega t + \frac{\phi}{2}\right)$$

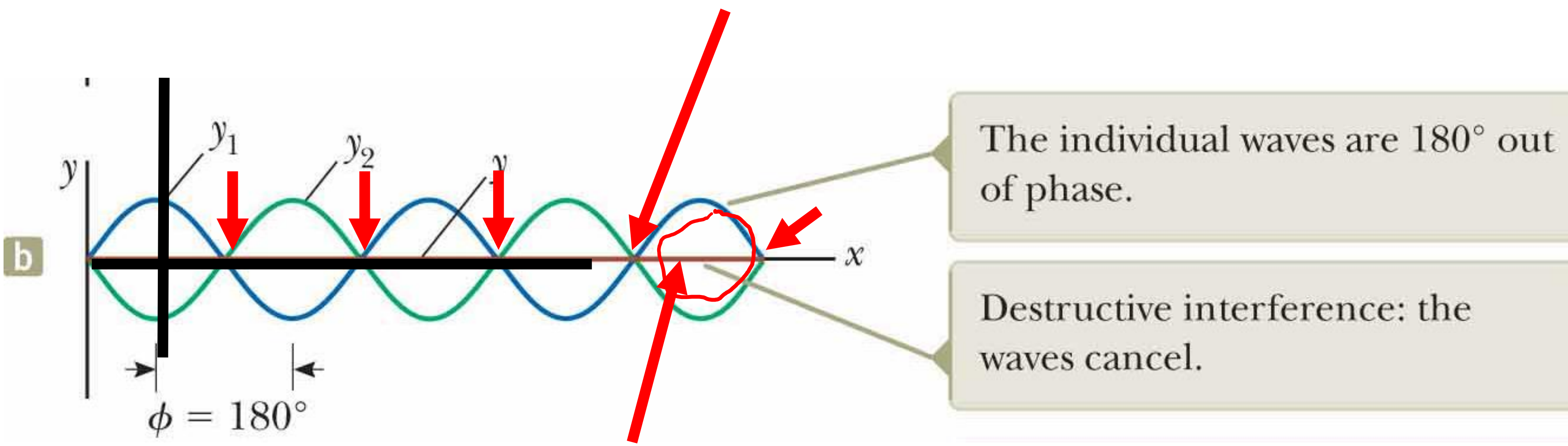
Path-length Difference:

Brown curve max at

$$\Delta r = \left|r_2 - r_1\right| = n\lambda, \quad n = 0, 1, 2, 3, \ldots$$

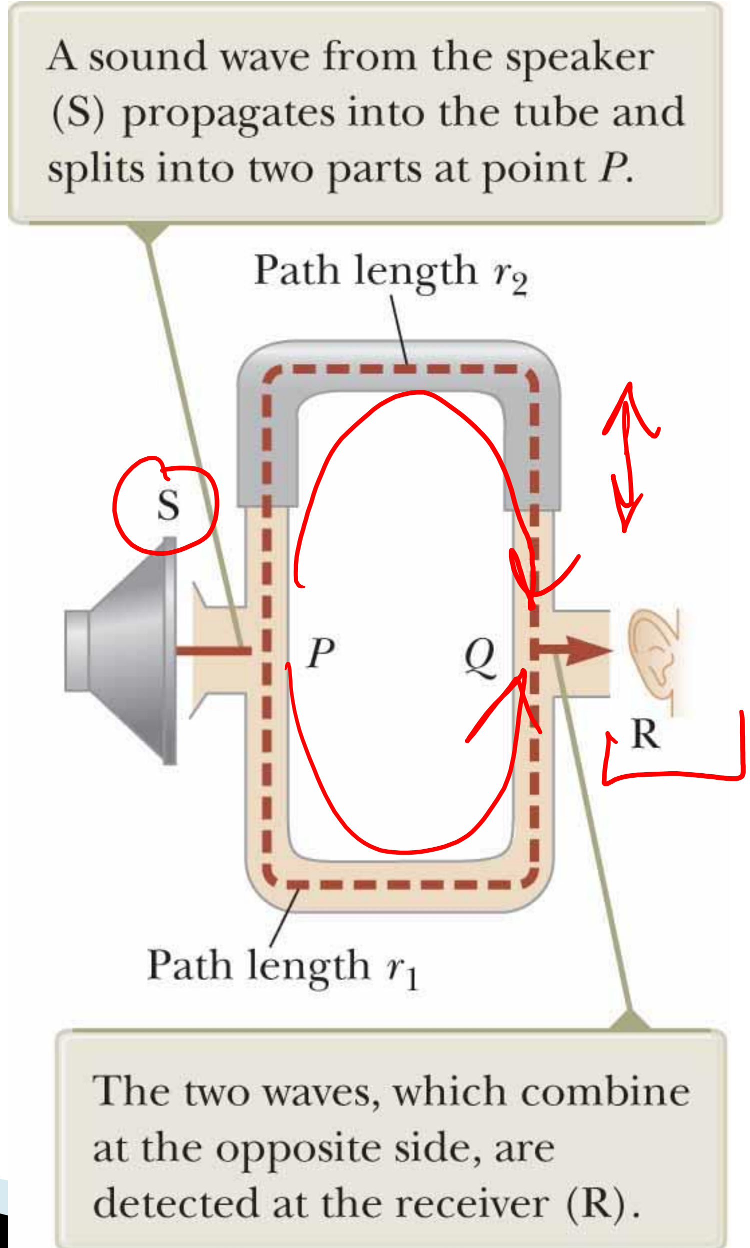


Superposition of Sinusoidal Waves



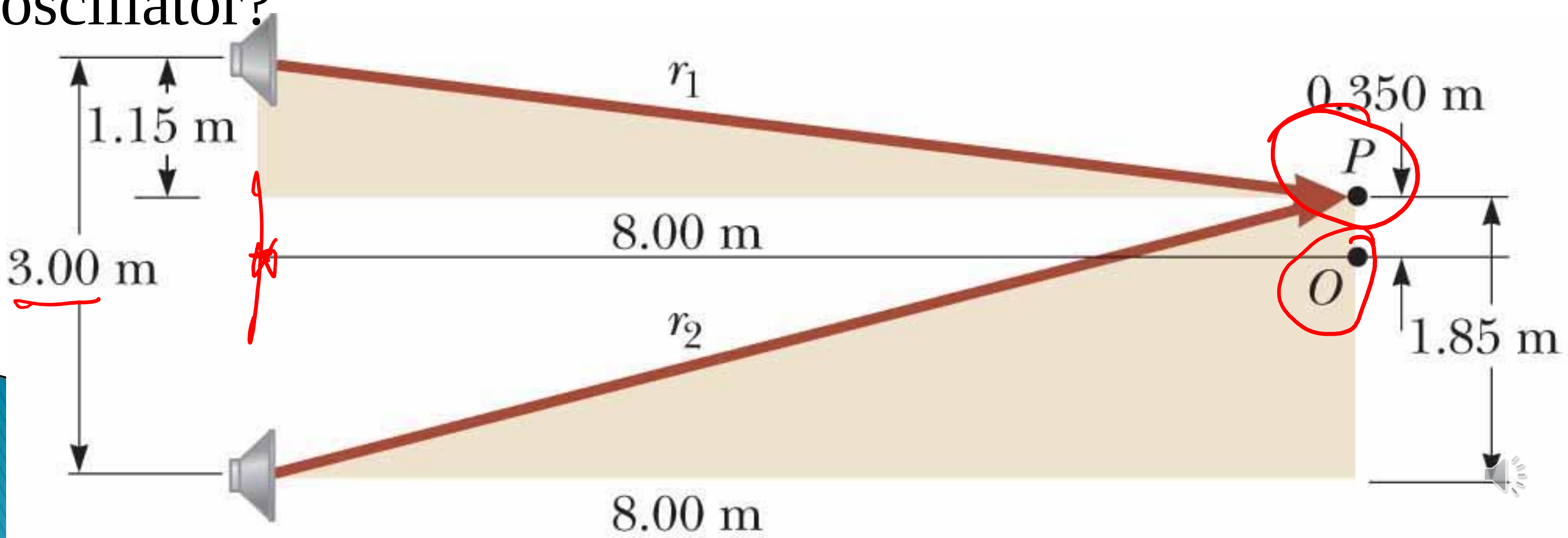
Brown curve = 0 everywhere

Superposition of Sinusoidal Waves



Example 17.1:
Two Speakers Driven by the Same Source

Two identical loudspeakers placed 3.00 m apart are driven by the same oscillator. A listener is originally at point O, located 8.00 m from the center of the line connecting the two speakers. The listener then moves to point P, which is a perpendicular distance 0.350 m from O, and she experiences the first minimum in sound intensity. What is the frequency of the oscillator?



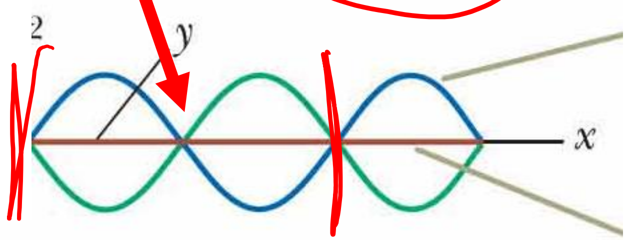
Example 17.1: Two Speakers Driven by the Same Source

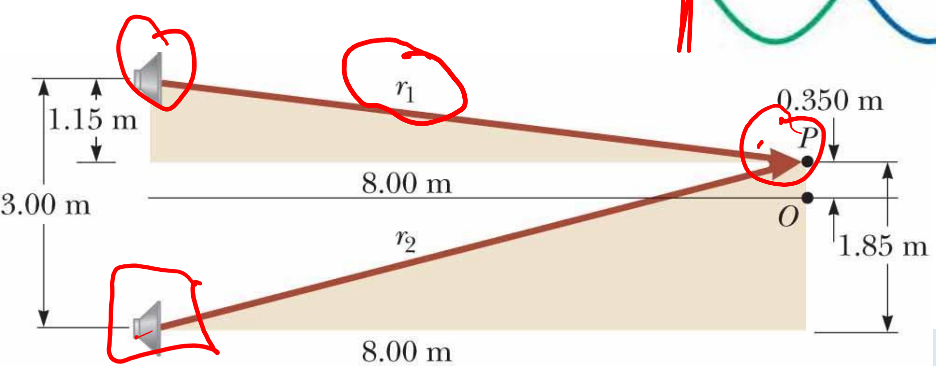
$$r_1 = \sqrt{(8.00\text{ m})^2 + (1.15\text{ m})^2} = 8.08\text{ m}$$

$$r_2 = \sqrt{(8.00\text{ m})^2 + (1.85\text{ m})^2} = 8.21\text{ m}$$

$$\text{For } r_2 - r_1 = 0.13\text{ m} = \frac{\lambda}{2} \qquad f = \frac{v}{\lambda} = \frac{343\text{ m/s}}{0.26\text{ m}} = \boxed{1.3\text{ kHz}}$$

$$\Rightarrow \lambda = 0.26\text{ m}$$





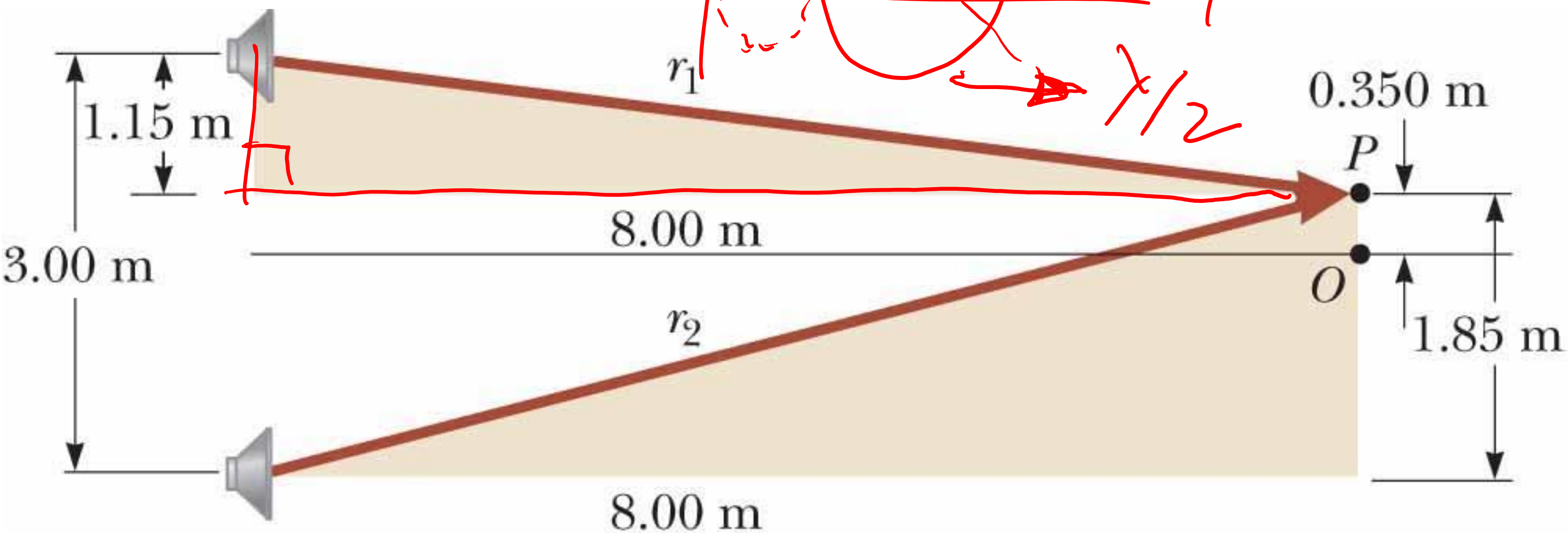
The first minimum occurs when the two waves reaching the listener at point P are 180° out of phase, in other words, when their path difference Δr equals $\lambda/2$.

Example 17.1: Two Speakers Driven by the Same Source

What if the speakers were connected out of phase? What happens at point *P*?

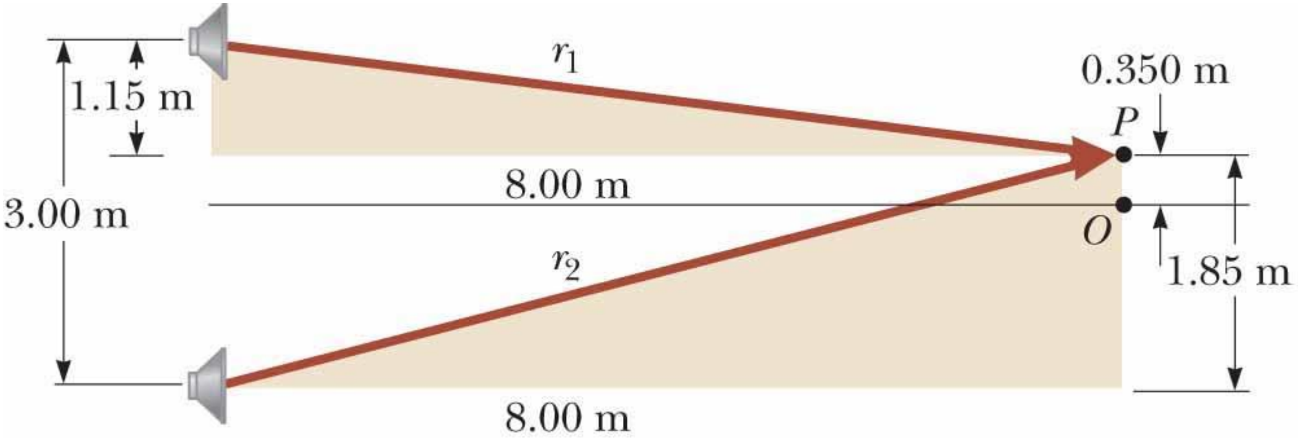
$\phi = 0$

$\lambda/2$



Example 17.1:
Two Speakers Driven by the Same Source

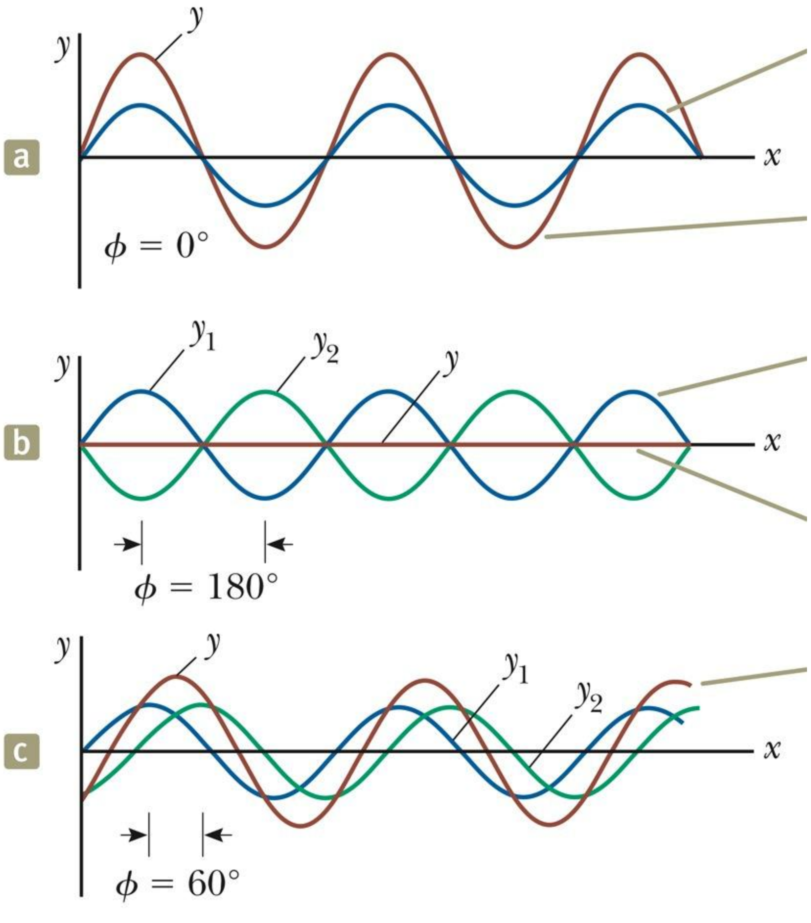
What if the speakers were connected out of phase? What happens at point *P*?



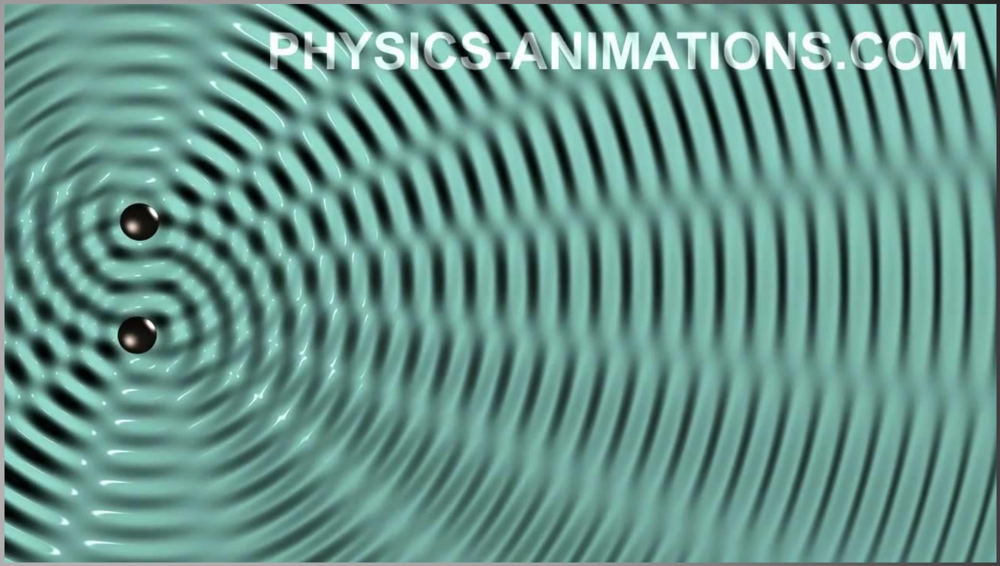
In this situation, the path difference of $\lambda/2$ combines with a phase difference of $\lambda/2$ due to the incorrect wiring to give a full phase difference of λ. As a result, the waves are in phase and there is a *maximum* intensity at point *P*.

Superposition of Sinusoidal waves

$$y' = \left[2A\cos\frac{\phi}{2}\right]\sin\left(kx - \omega t + \frac{\phi}{2}\right)$$



- If $\phi = 0$, (waves are in phase) then amplitude is $2A$
  - Also when $\phi = 2n\pi$, $n = 0, 1, 2, 3$ ….
  - Constructive interference
- If $\phi = \pi$, (waves are out of phase) then amplitude is 0
  - Also when $\phi = n\pi$, $n = 1, 3, 5$ ….
  - Destructive interference
- General case: same λ, but different A.
  - If $\phi = 60°$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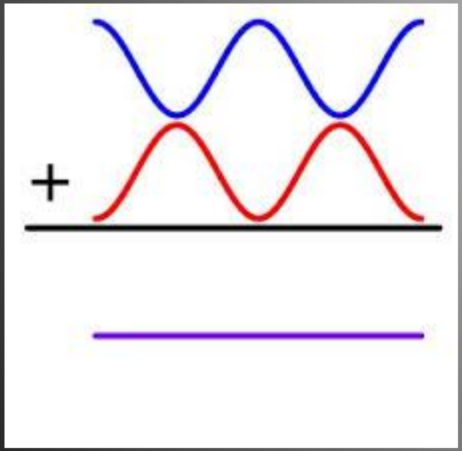
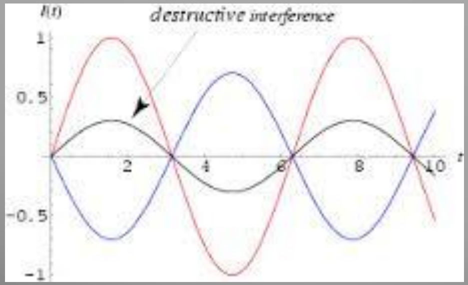
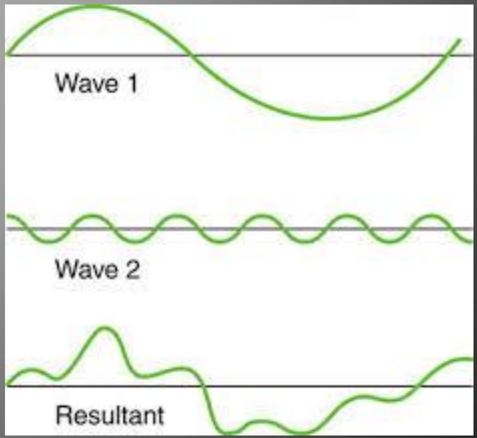

+


destructive interference
1
0.5
-0.5
-1
2
4
6
8
10


Wave 1
Wave 2
Resultant

Standing waves

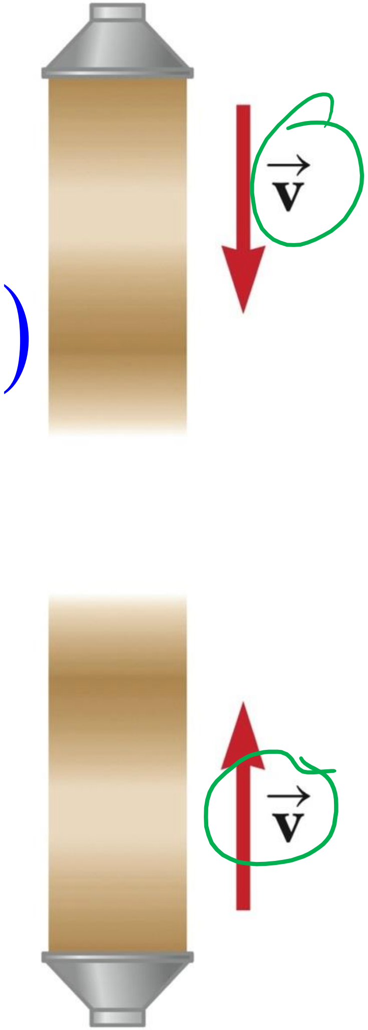
- Previous section – waves travelling in the same direction.
- This section – waves travel in opposite directions
- $y_1 = A\sin\left(kx - \omega t\right) \qquad y_2 = A\sin\left(kx + \omega t\right)$
- Resultant wave function:

$$y'(x,t) = y_1(x,t) + y_2(x,t)$$

$$= A\sin(kx - \omega t) + A\sin(kx + \omega t)$$

Since $\sin\left(a \pm b\right) = \sin a \cos b \pm \cos a \sin b$

- Then $y'\left(x,t\right) = \underbrace{(2A\ \sin kx)}_{\text{amplitude}}\ \cos \omega t$

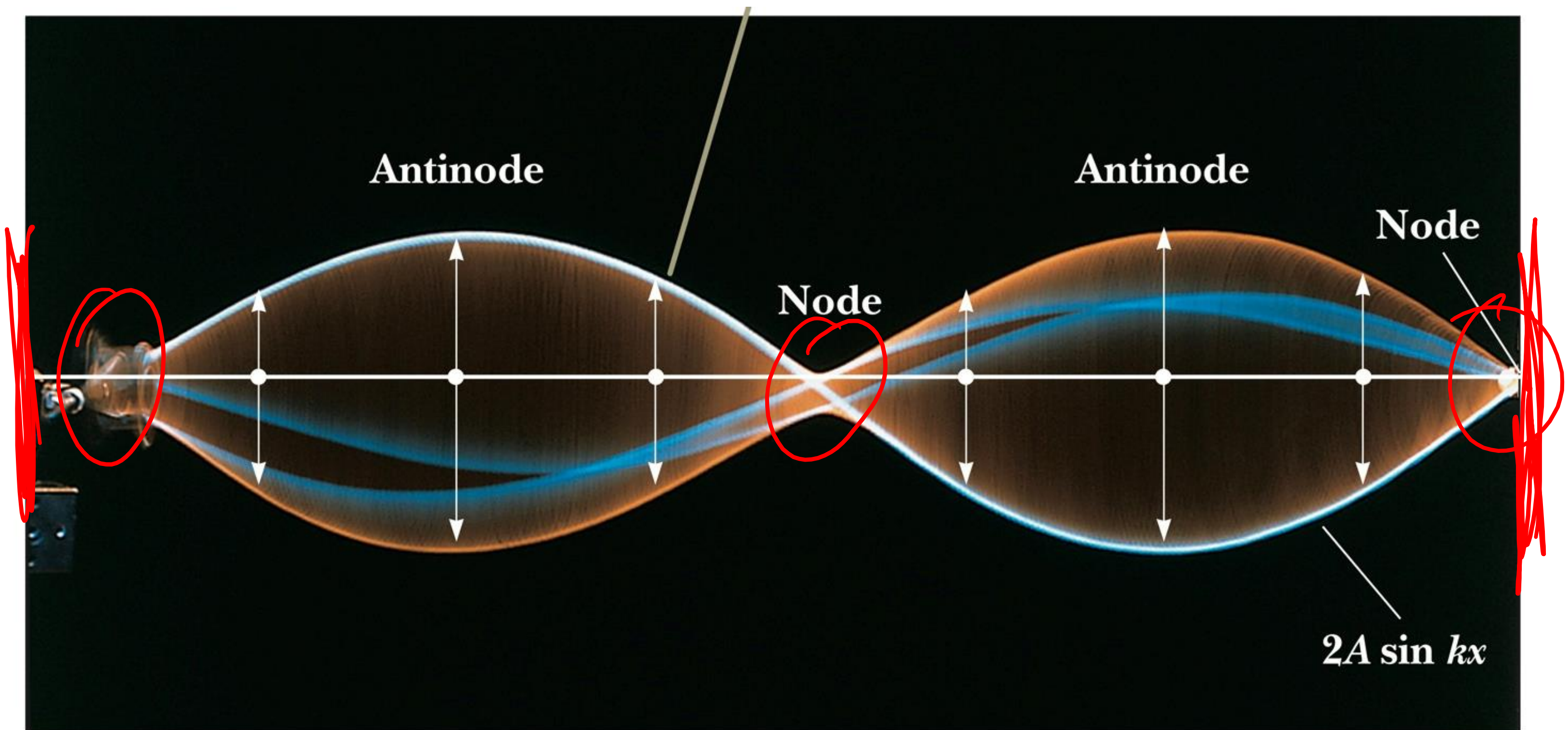


- **Standing waves**
  Two sinusoidal waves of same amplitude and wavelength travel in opposite directions along a stretched string, their interference produces a standing wave.

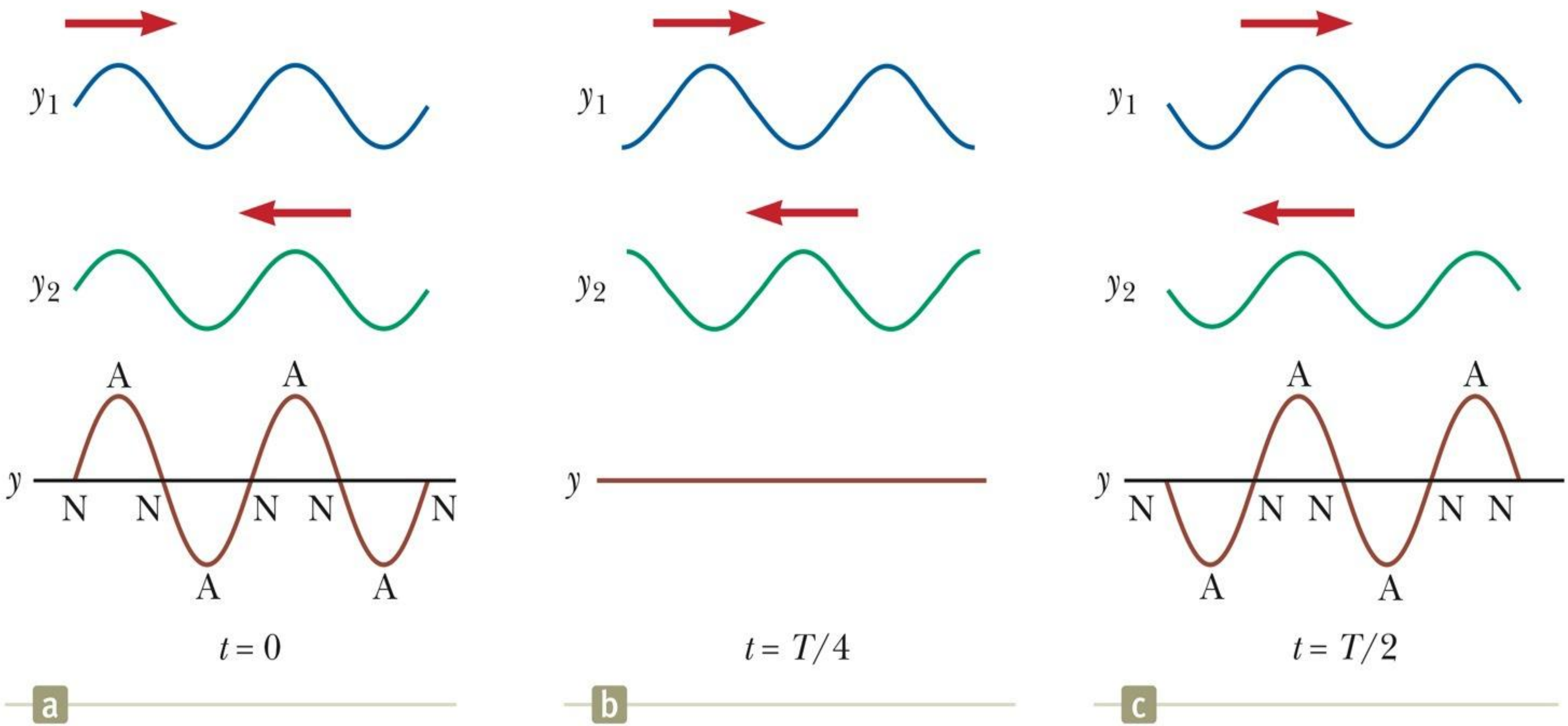
Standing waves

$$y'(x,t) = (2A\sin kx)\ \cos\omega t$$

- Standing waves
  Two sinusoidal waves of same amplitude and wavelength travel in opposite directions along a stretched string, their interference produces a standing wave.
- There is no (kx-ωt) in the function – not an expression for a single wave.



Standing waves



Standing Waves –2 Fixed Ends

$$y'(x,t) = (2y_m \sin kx)\cos \omega t$$

- Amplitude is zero where ($\sin kx$) = 0

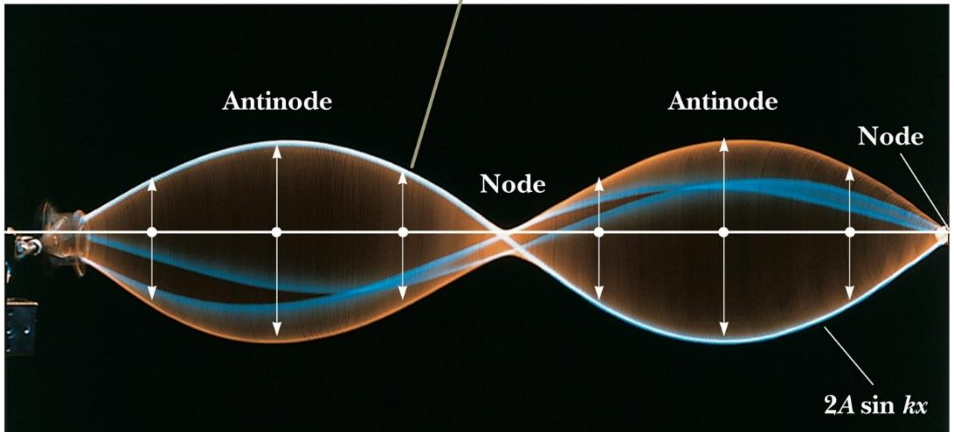
  $kx = n\pi$ for n = 0, 1, 2, 3, …

  substitute $k = 2\pi/\lambda$ (wave number)

$(2\pi/\lambda)x = n\pi$ *implies* $x = n\pi \dfrac{\lambda}{2\pi} = n\dfrac{\lambda}{2}$

for n = 0, 1, 2, …

(nodes – zero amplitude)



Standing Waves

$$y'(x,t) = (2y_m \sin kx)\cos\omega t$$

Amplitude maximum where $|\sin kx| = 1$

$kx = (n+½)\pi$ for $n = 0, 1, 2, \ldots$

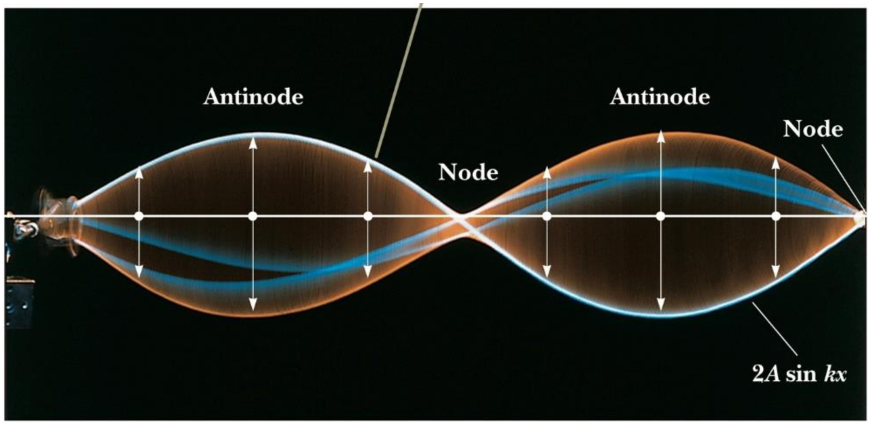
Substituting $k = 2\pi/\lambda$ in $|\sin kx| = 1$

$$kx = (n+\frac{1}{2})\pi$$

$$\left(\frac{2\pi}{\lambda}\right)x = \left(n+\frac{1}{2}\right)\pi$$

$$x = \frac{\lambda}{2\pi}\left(n+\frac{1}{2}\right)\pi$$

$$= \frac{\lambda}{2}\left(n+\frac{1}{2}\right) \qquad n = 0,1,2,3,\ldots$$



(Try it out for a few n values)

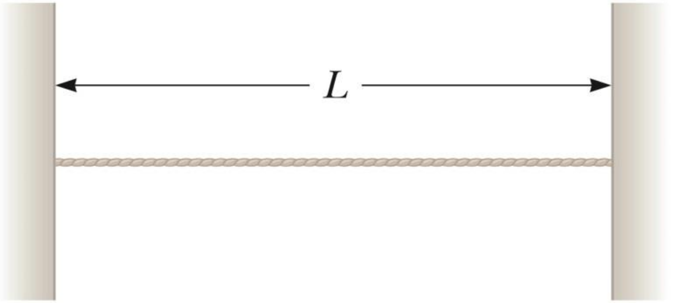
$$= n\frac{\lambda}{4} \qquad n = 1,3,5,\ldots$$

(antinodes - points of max amplitudes)

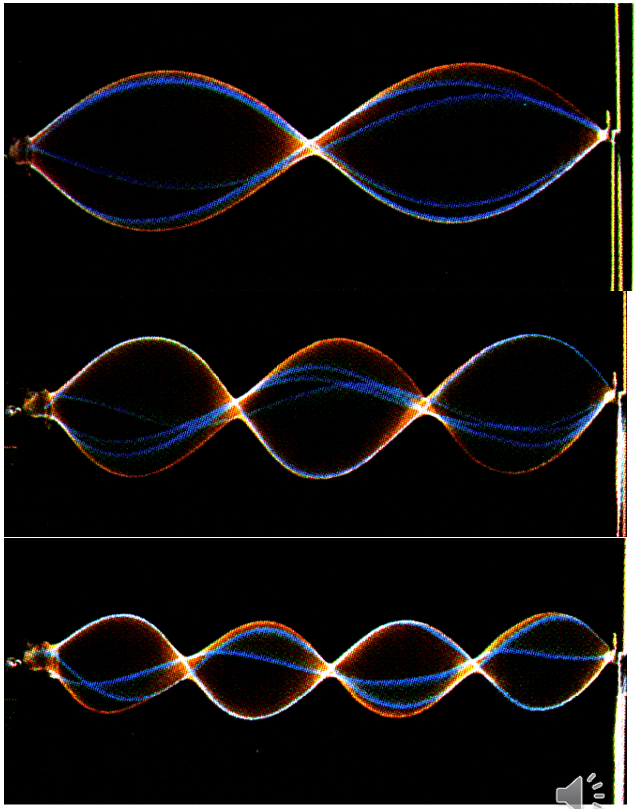
Standing waves and Resonance

Miscellaneous>Videos>>Standing Waves Part I & II:

Standing Waves Part I: Demonstration
Standing Waves Part II: Explanation

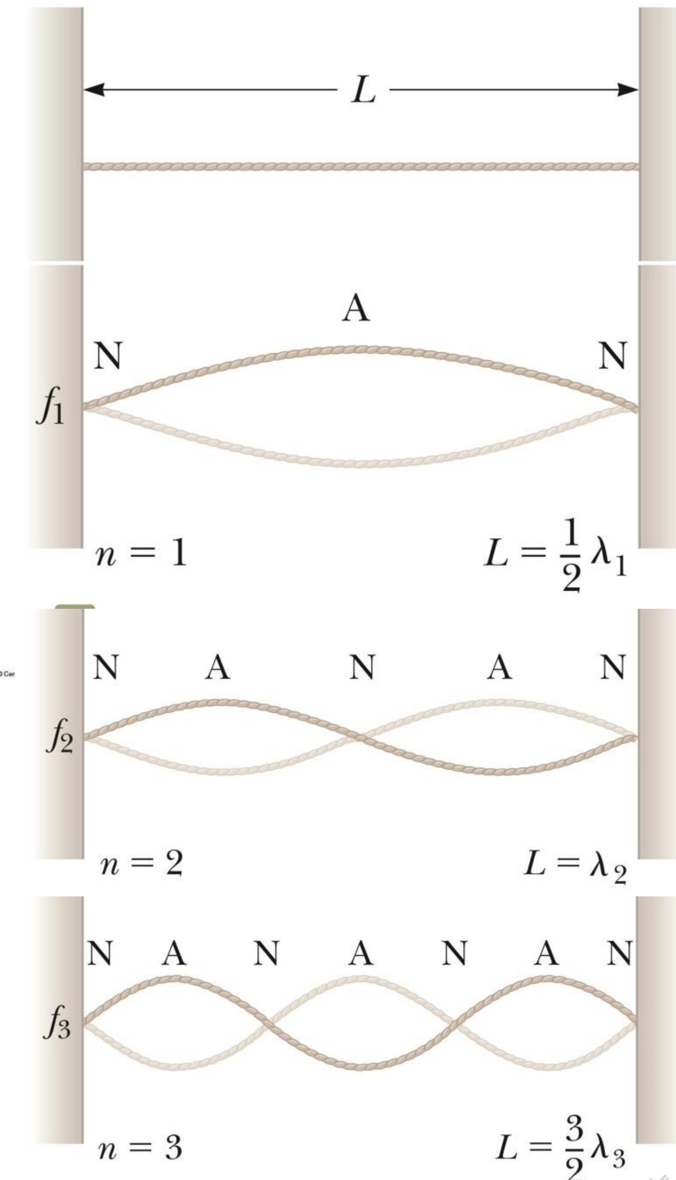


- Consider a string, stretched between 2 clamps a distance L apart, e.g. guitar.
- Send a continuous, sinusoidal wave toward right.
- Boundary conditions: $y = 0$ at $x = 0$ and $y = 0$ at L
- The continuous wave Reflects back and travels toward the left.
- When reaching the left side, reflects back…



Standing waves & bounary conditions

- For specific frequencies – standing waves are set up with nodes and anti-nodes.
- Quantisation.
- Strings will resonate at resonant frequencies.
- When not at resonant frequency *f*, no standing waves will form.



Standing waves and Resonance

- Two nodes, one antinode

$$\lambda = 2L$$

- Three nodes, two antinodes:

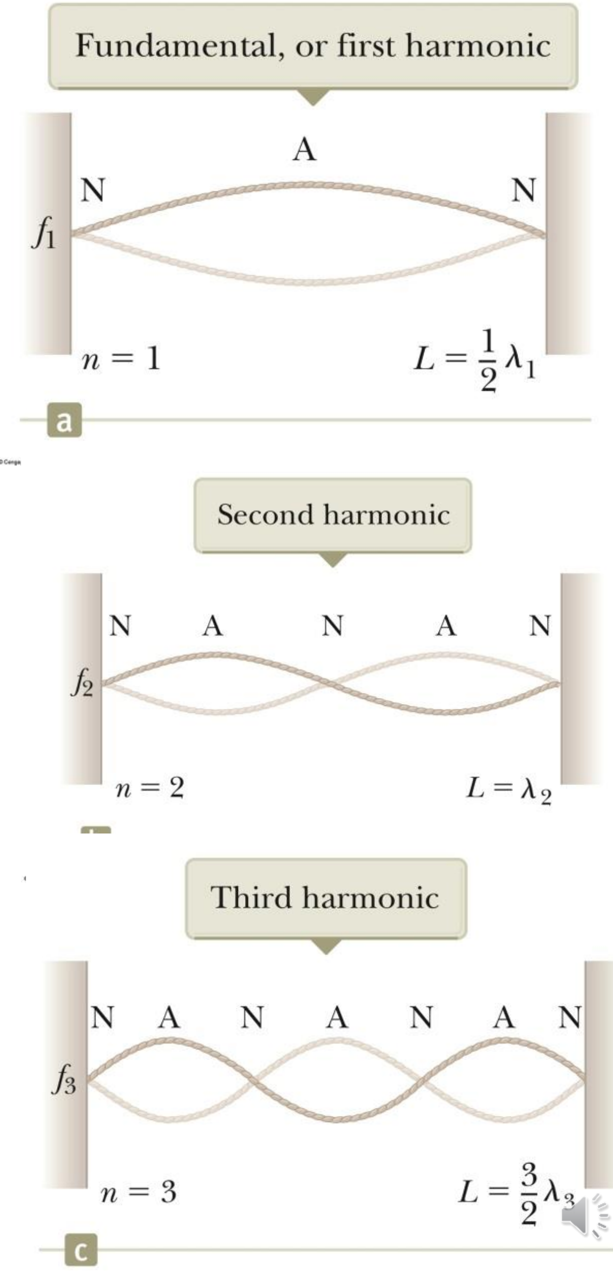
$$\lambda = L$$

- Four nodes, three antinodes

$$\lambda = \frac{2}{3}L$$

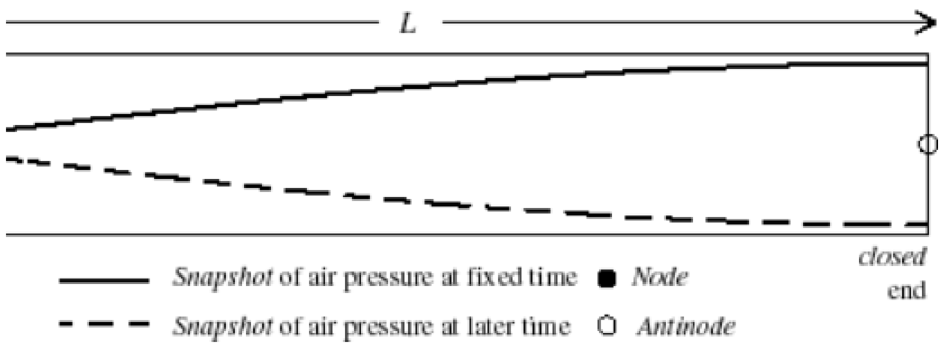
- Relation between λ and L for standing wave

$$\lambda = \frac{2L}{n}, \qquad \text{for } n = 1, 2, 3, \ldots$$

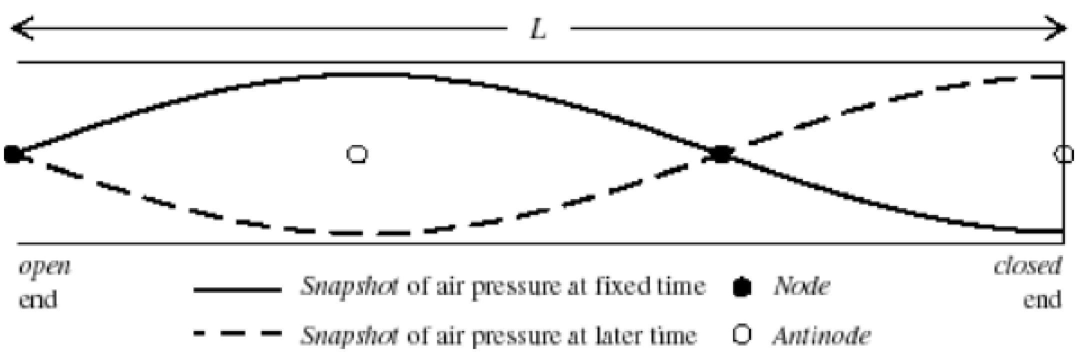


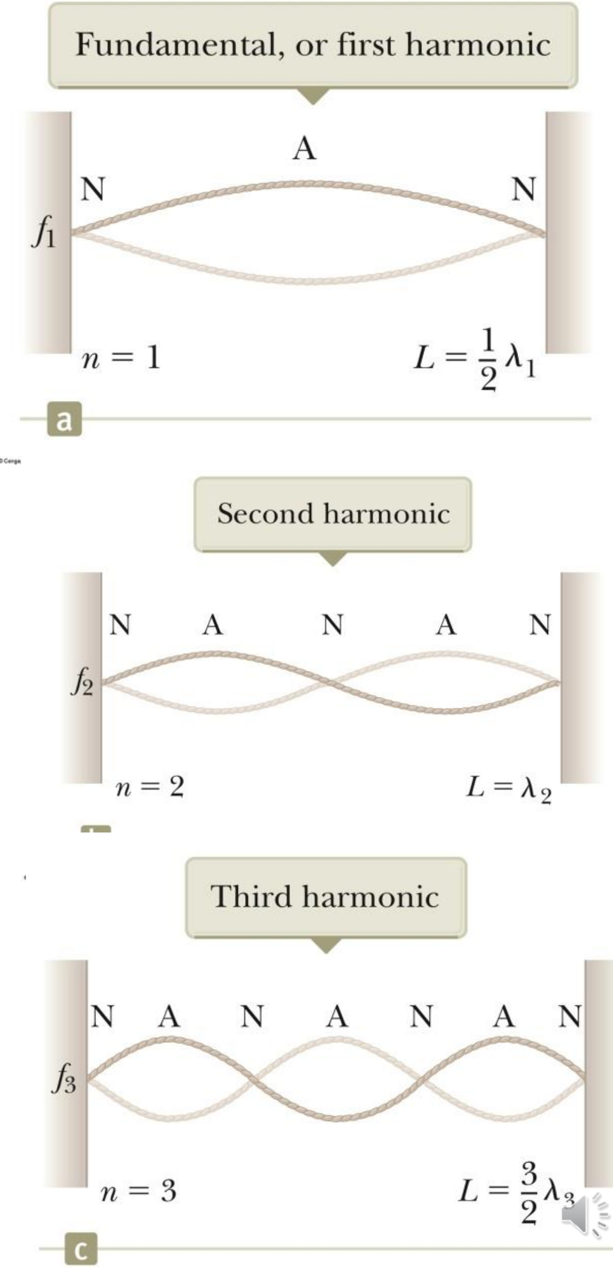
Standing waves and Resonance

Fundamental Mode: Standing Sound Wave in a Pipe



Second Harmonic: Standing Sound Wave in a Pipe





Standing waves and Resonance

- In general resonant frequencies/Harmonics:

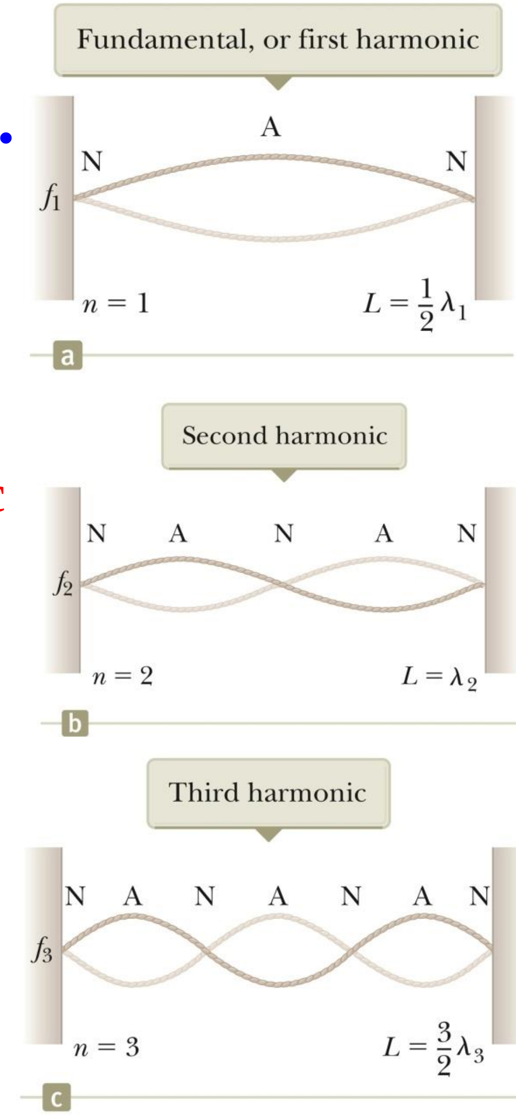
$$\lambda = \frac{2L}{n}, \qquad \text{for } n = 1, 2, 3, \ldots$$

$$f = \frac{v}{\lambda} = n\frac{v}{2L}, \qquad \text{for } n = 1,\ 2,\ 3, \ldots.$$

For $n = 1$: $f_1 = v/2L$ Fundamental mode or first harmonic

For $n = 2$: $f_2 = v/L$ Second harmonic

For $n = 3$: $f_3 = 3v/2L$ Third harmonic



Problem

- Standing waves can be excited on a string of length $L = 1.0$ m whose ends are fixed at $x = 0$ and $x = L$. Find the 2 longest wavelengths and 2 smallest values of the wave number k.
- Wavelength :
- Wave number:

Problem (Self Study)

- Standing waves can be excited on a string of length $L = 1.0$ m whose ends are fixed at $x = 0$ and $x = L$. Find the 2 longest wavelengths and 2 smallest values of the wave number k.
- Wavelength :

$$\lambda = \frac{2L}{n}, \qquad \text{for } n = 1, 2, 3, \ldots$$

$$\boxed{\lambda = \frac{2L}{n}}$$

$$n = 1: \lambda = 2L = 2.0 \text{ m}$$

$$n = 2: \lambda = L = 1.0 \text{ m}$$

- Wave number:

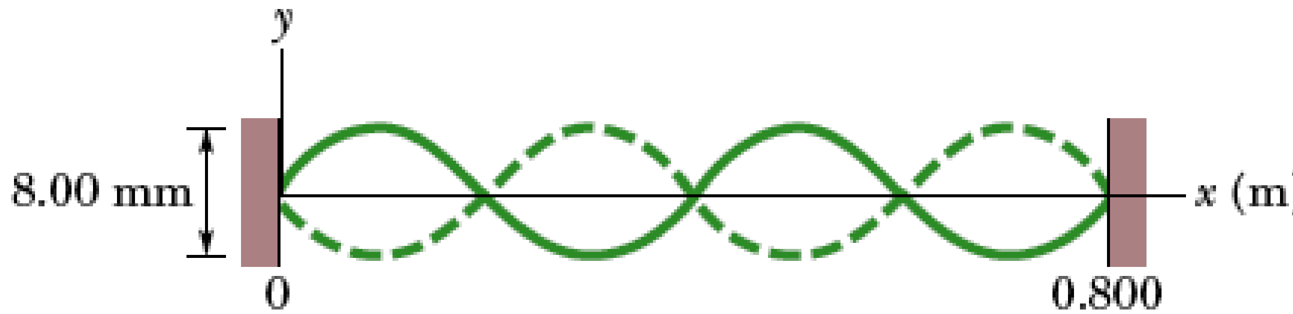
$$\boxed{k = \frac{2\pi}{\lambda} = \frac{\pi n}{L}}$$

$$n = 1: k = \pi \text{ m}^{-1}$$

$$n = 2: k = 2\pi \text{ m}^{-1}$$

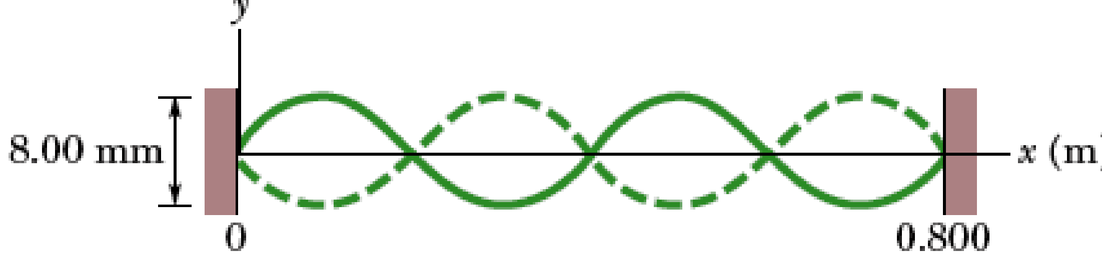
Problem

$v = \sqrt{\frac{T}{\mu}}$



- The figure shows a pattern of resonant oscillation of a string of mass m = 2.500 g and length L = 0.800 m that is under tension T = 325.0 N. The wave has a velocity of 322 m/s.
- What is the wavelength λ of the transverse waves producing the standing wave pattern, and what is the harmonic number n?
- What is the frequency f of the transverse waves?
- What is the maximum magnitude of the transverse velocity v_m of the element oscillating at coordinate x = 0.180 m.
- At what point during the element's oscillation is the transverse velocity maximum?

Problem



The figure shows a pattern of resonant oscillation of a string of mass $m = 2.500$ g and length $L = 0.800$ m that is under tension $T = 325.0$ N. The wave has a velocity of 322 m/s. What is the wavelength λ of the transverse waves producing the standing wave pattern, and what is the harmonic number n? What is the frequency f of the transverse waves? What is the maximum magnitude of the transverse velocity v_m of the element oscillating at coordinate $x = 0.180$ m. At what point during the element's oscillation is the transverse velocity maximum?

From the figure,
$2\lambda = L$
$\lambda = 0.400$ m

$$\lambda = \frac{2L}{n}, \quad \text{for } n = 1, 2, 3, \ldots$$

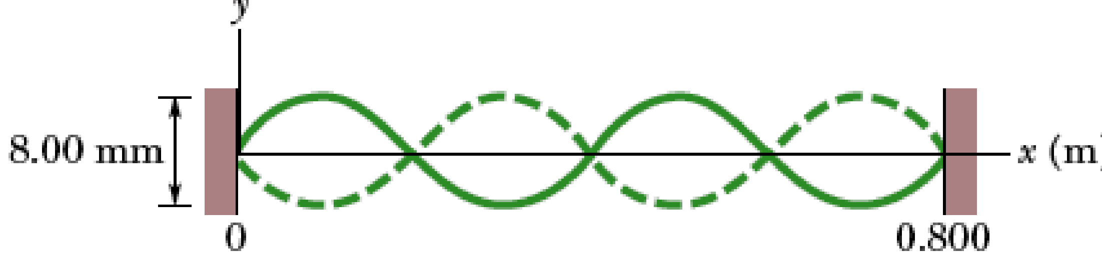
$$\lambda = \frac{2L}{n}$$

$$n = \frac{2L}{\lambda} = \frac{2(0.800)\text{ m}}{0.400\text{ m}} = 4$$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{325.0\text{ N}}{(2.500\text{ g}/0.800\text{ m})}} = 322.49\text{ m/s}$$

$$f = \frac{v}{\lambda} = \frac{322\text{ m/s}}{0.400\text{ m}}\pi$$
$$= 806.225\text{ Hz}$$

Problem



From calculations

$$\lambda = 0.400 \text{ m}$$
$$n = 4$$
$$f = 806.225 \text{ Hz}$$

$$y'(x,t) = (2A\sin kx)\cos\omega t$$

The maximum amplitude occurs when sin kx = ±1

i..e. $2A = 4.00 \text{ mm} \;\; \therefore A = 2.00 \text{ mm}$

$$v_{\text{transverse}} = \frac{\partial y'}{\partial t} = (-2\omega A\sin kx)\sin\omega t$$

$$\left|v_{\text{y,transverse}}\right| = 2\omega A\sin kx$$

$$\therefore v_{\text{y,transverse}}(x = 0.180 \text{ m}) =$$

$$2(2\pi f)A\sin\left(\left(\frac{2\pi}{\lambda}\right)x\right)$$

$$= 2(2\pi \times 806.25 \text{ Hz})(2.00\times10^{-3} \text{ m})\sin\left(\left(\frac{2\pi}{0.400 \text{ m}}\right)0.180 \text{ m}\right)$$

$$= 6.26 \text{ m/s}$$

Maximum speed occurs at equilibrium pos. Like in the spring-block SHM case.